

Regularized Linear Regression

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EVERY GROUP PROJECT



**DOES 99%
OF THE WORK**

**HAS NO IDEA
WHAT'S GOING
ON THE
WHOLE TIME**

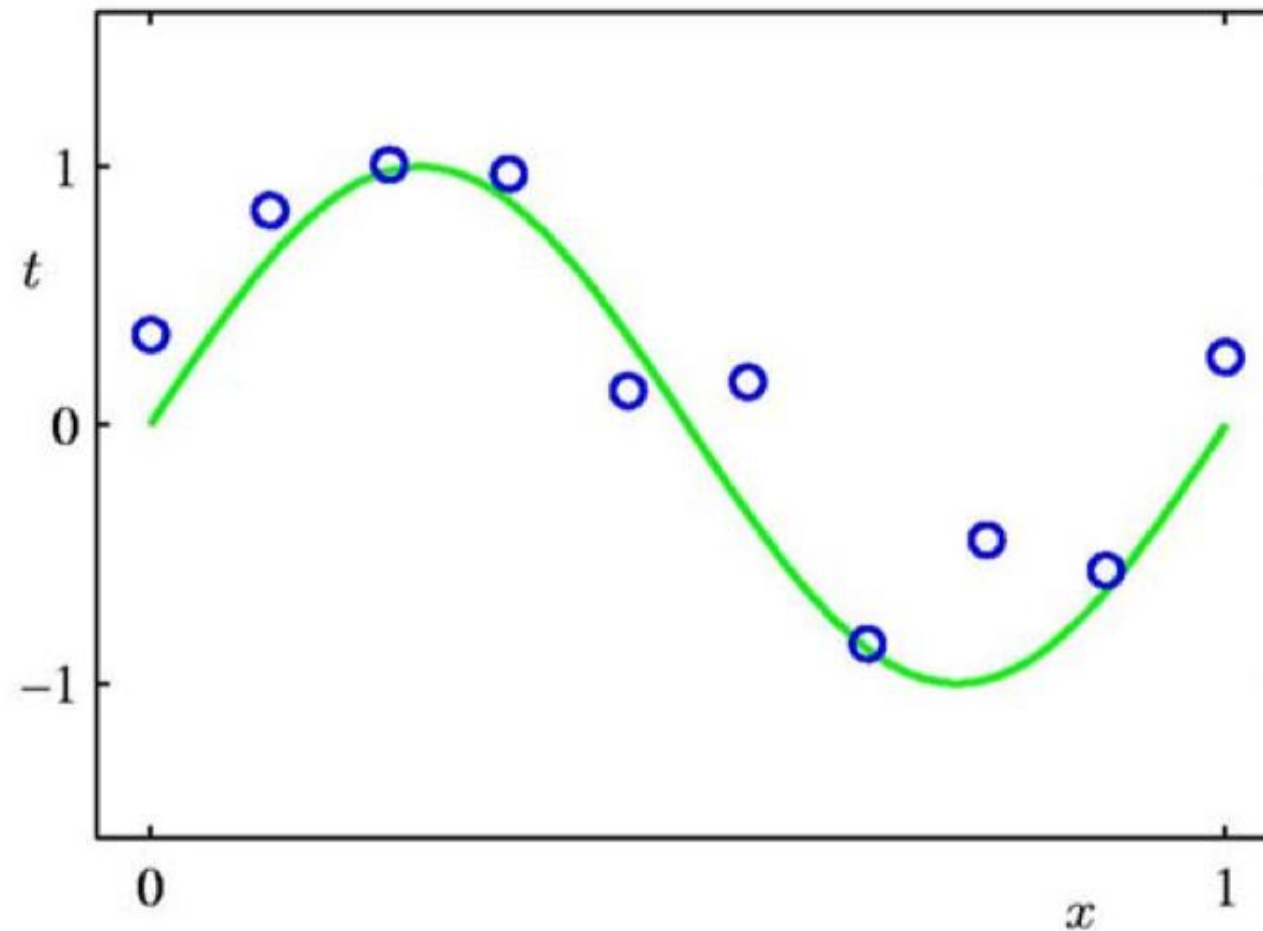
**SAYS HE'S
GOING TO
HELP
BUT HE'S
NOT**

**DISAPPEAR
AT THE VERY
BEGINNING AND
DOESN'T SHOW
UP AGAIN TIL
THE VERY END**

Outline

- Overfitting and regularized learning 
- Ridge regression
- Lasso regression
- Determining regularization strength

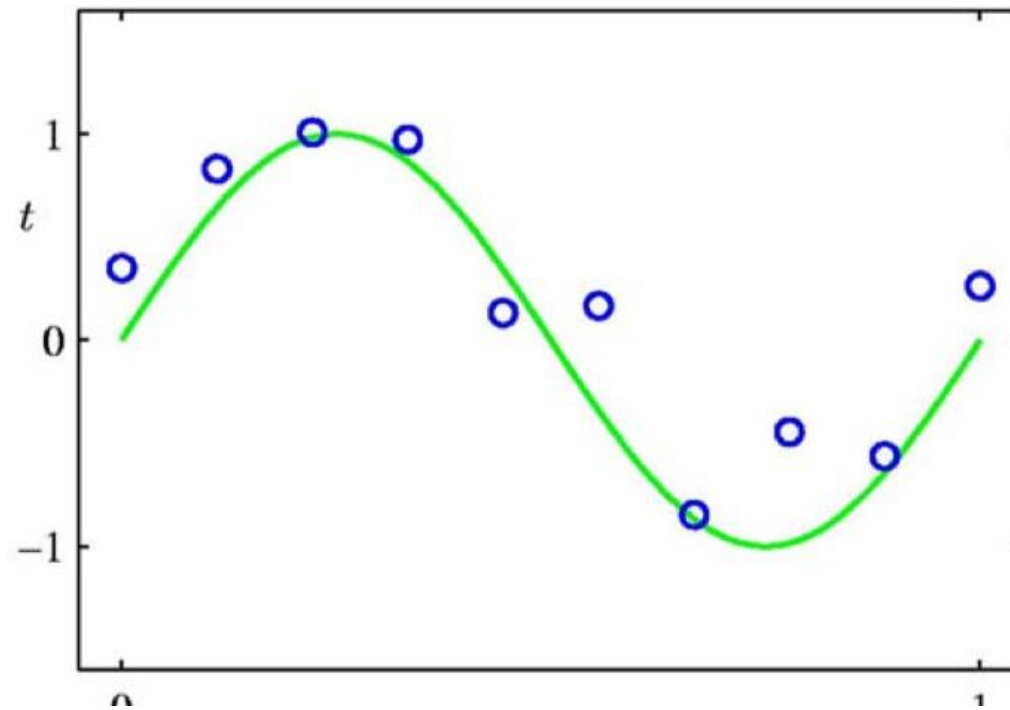
Regression: Recap



- Suppose we are given a training set of N observations
- Regression problem is to estimate $y(x)$ from this data

Regression: Recap

$$x = \begin{bmatrix} h \end{bmatrix}$$



$$z = \begin{bmatrix} h & h^2 & h^3 & \dots & h^d \end{bmatrix}$$

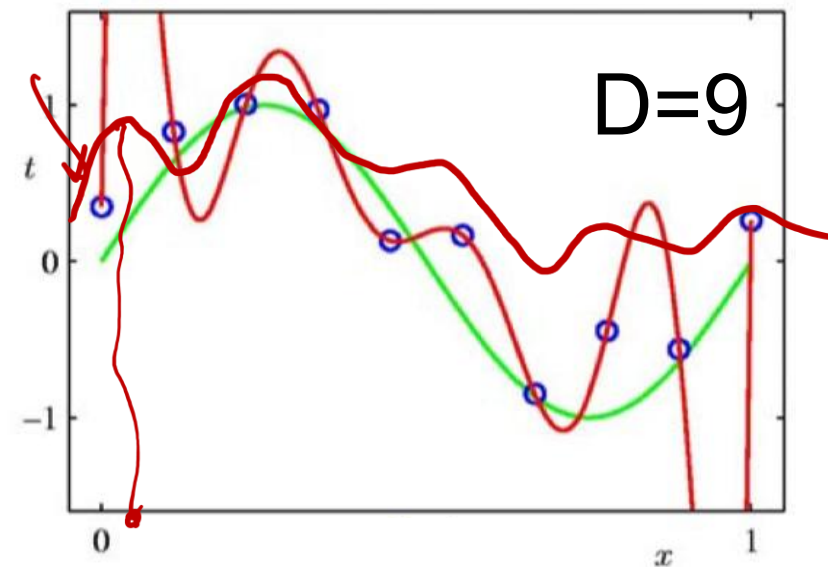
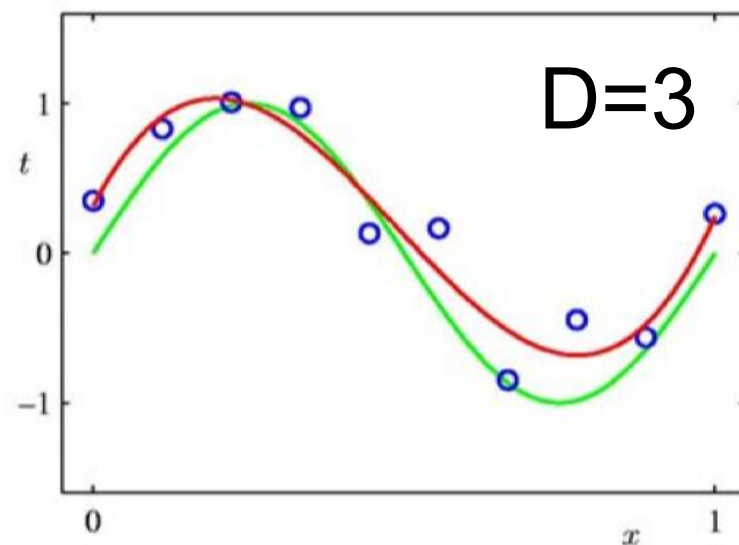
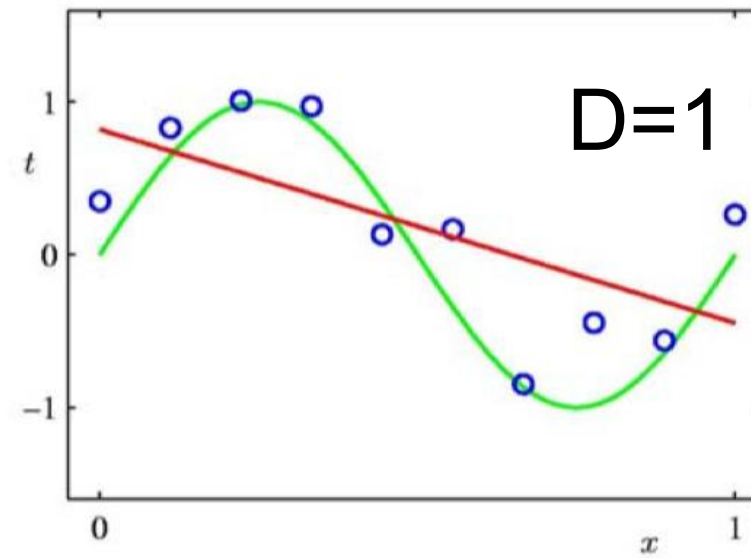
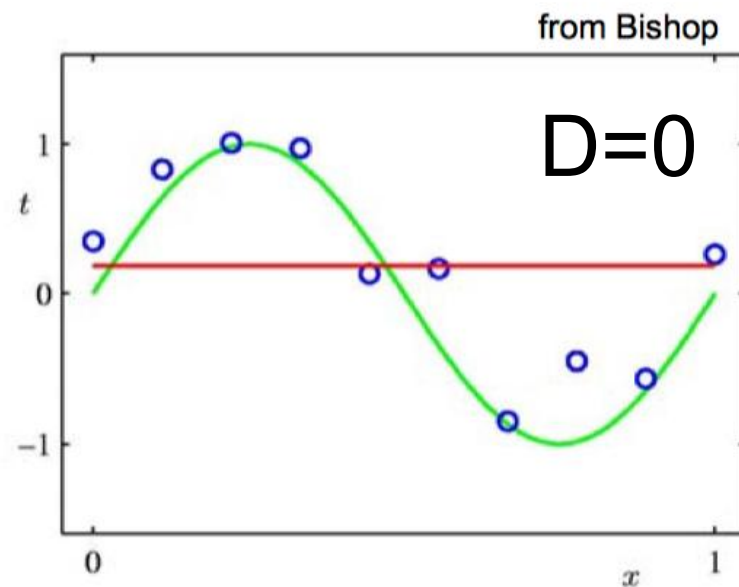
- Want to fit a polynomial regression model

$$y = \theta_0 + \theta_1 x + \theta_2 x^2 + \dots + \theta_d x^d + \epsilon$$

- $z = \{1, x, x^2, \dots, x^d\} \in R^d$ and $\theta = (\theta_0, \theta_1, \theta_2, \dots, \theta_d)^T$

$$y = z\theta$$

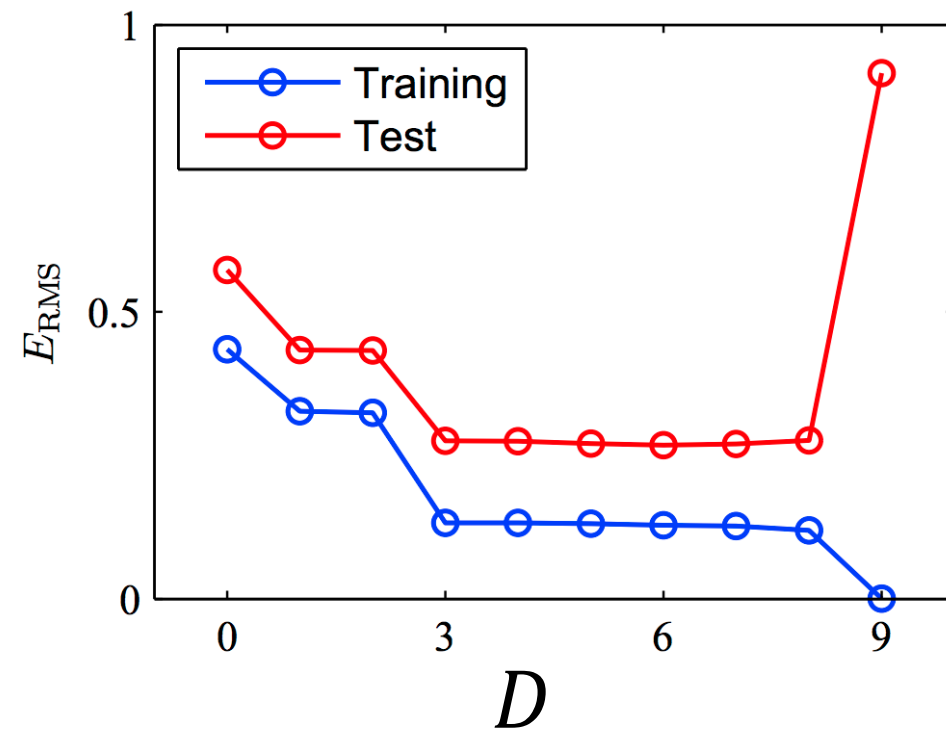
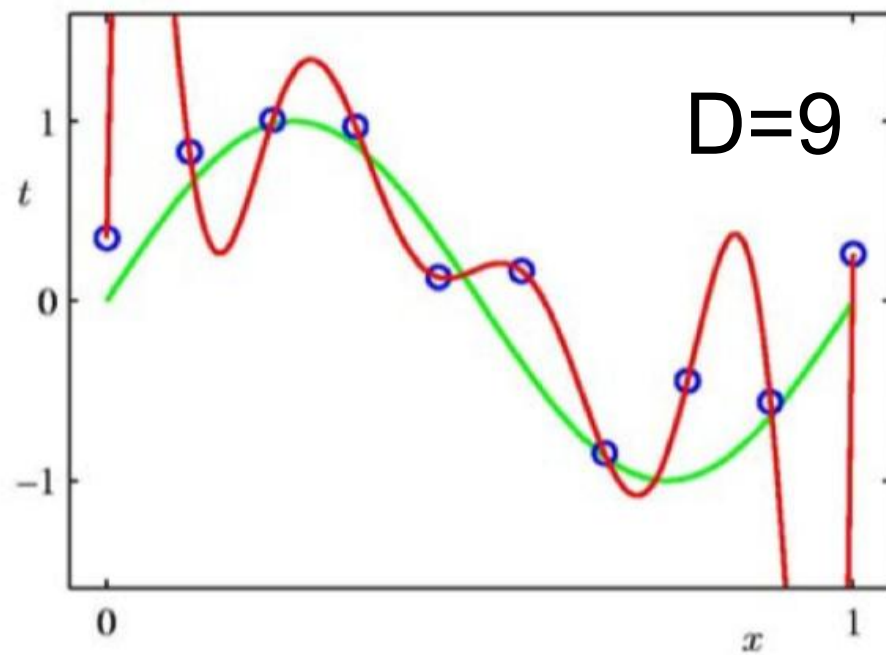
Which One is Better?



- Can we increase the maximal polynomial degree to very large, such that the curve passes through all training points?

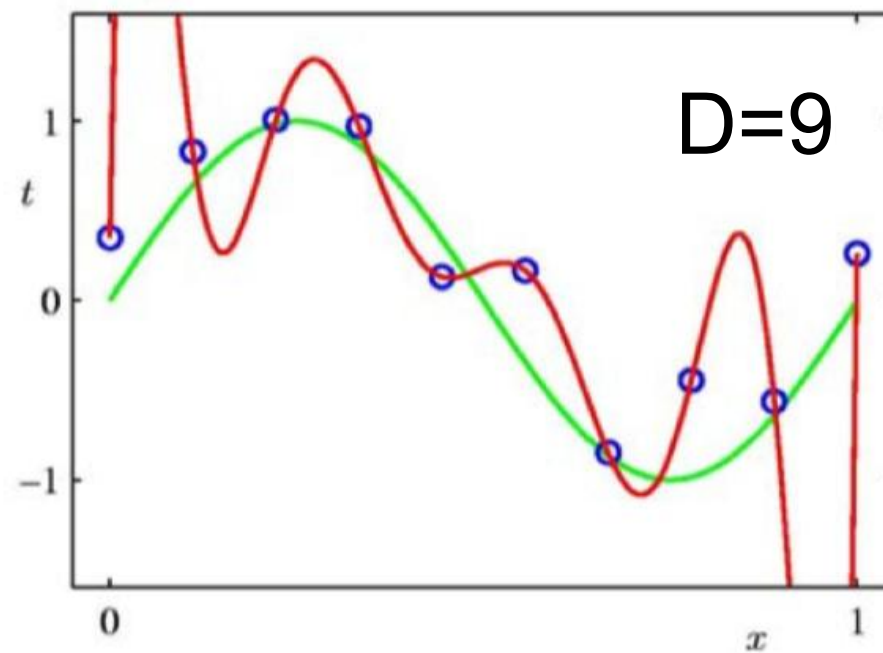
No, this can lead to **overfitting**!

The Overfitting Problem



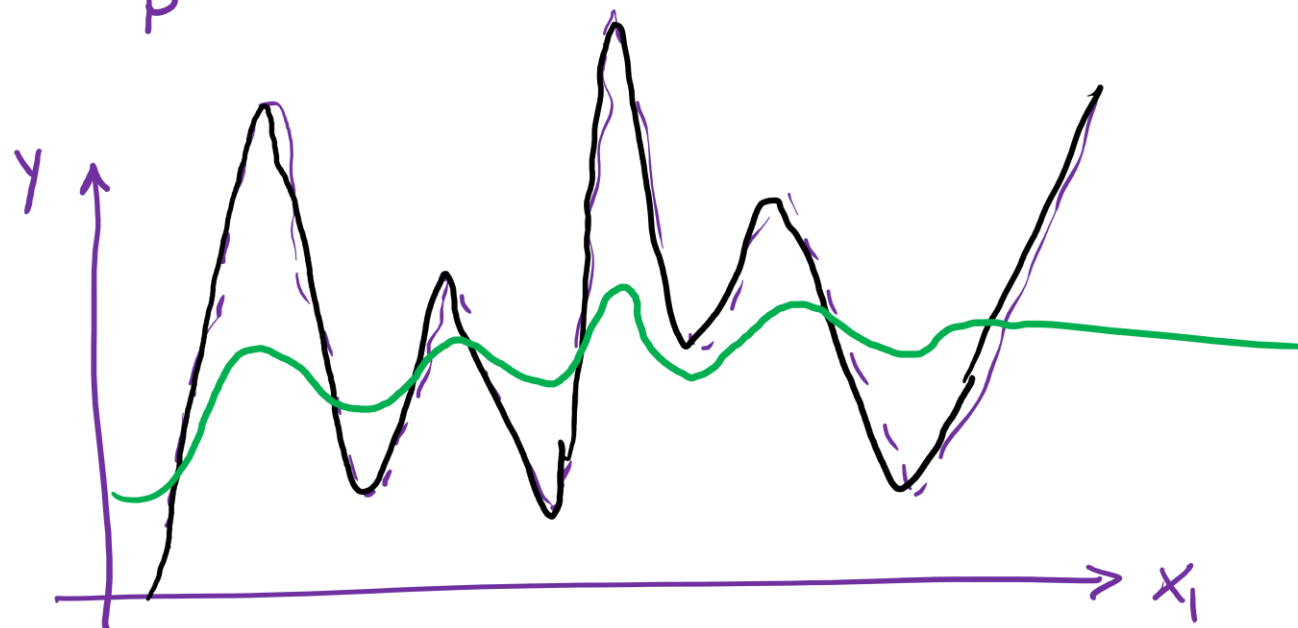
- The training error is very low, but the error on test set is large.
- The model captures not only patterns but also noisy nuisances in the training data.

The Overfitting Problem



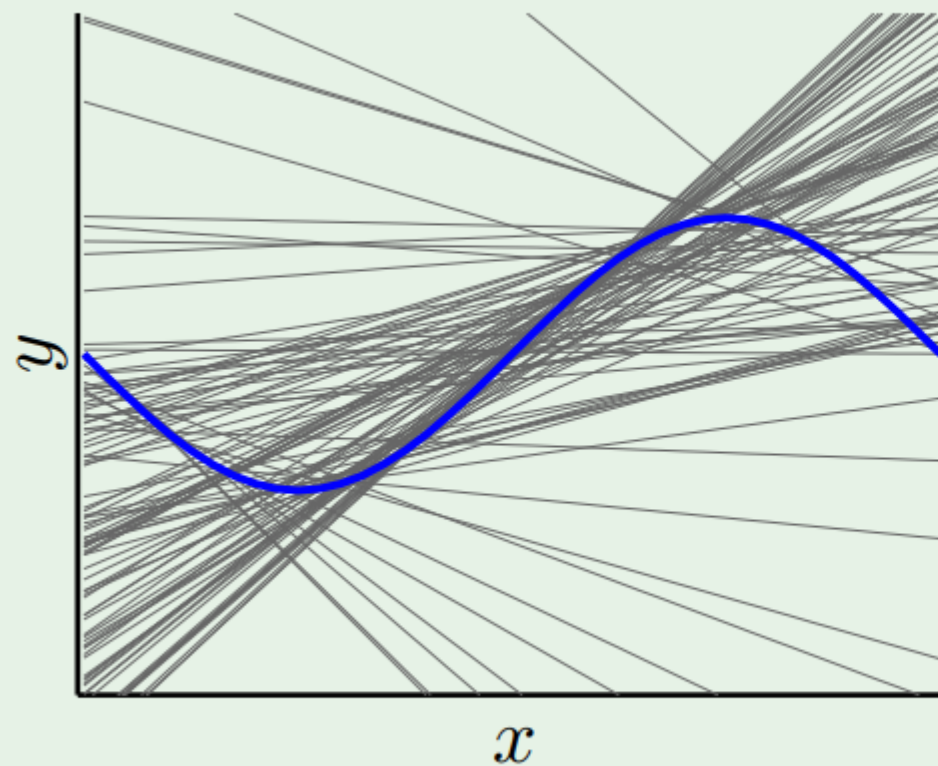
- In regression, overfitting is often associated with large Weights (**severe oscillation**)
- How can we address overfitting?

$$\hat{y}_p = \theta_0 + \theta_1 x_1 + \theta_2 x_1^2 + \dots + \theta_d x_1^d$$

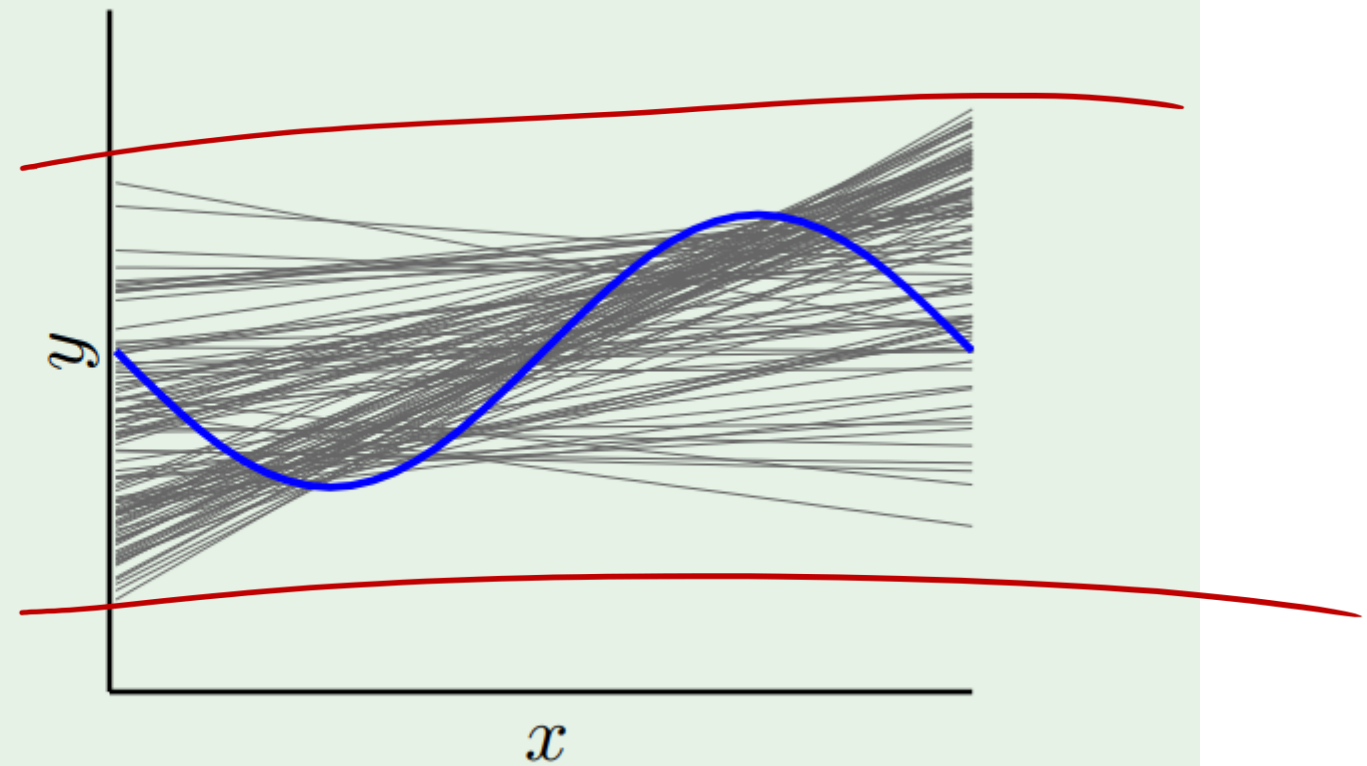


Regularization

(smart way to cure overfitting disease)



without regularization



with regularization

Put a brake on fitting



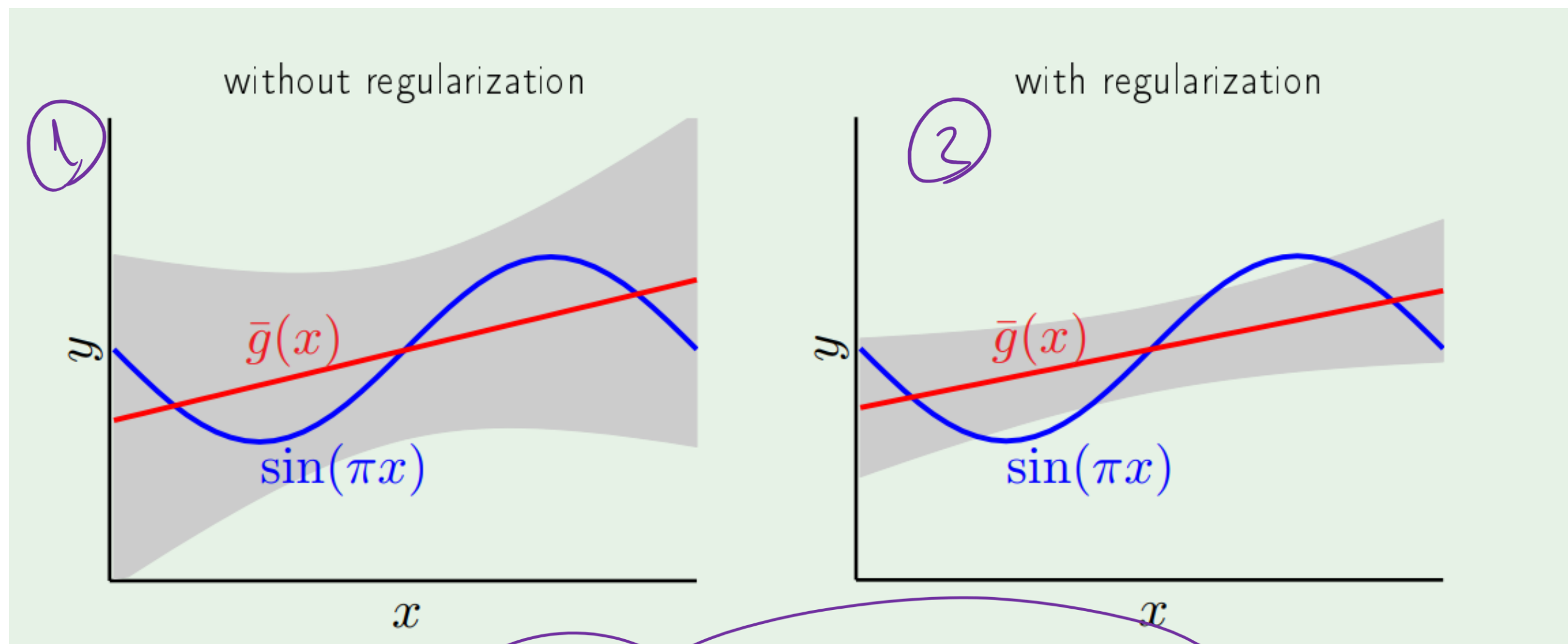
Fit a linear line on sinusoidal with just two data points

$$\hat{y}_p^{(i)} = z^{(i)} \Theta$$

Who is the winner?

$$E(\Theta) = \frac{1}{N} \sum_{i=1}^N (y_a^{(i)} - \hat{y}_p^{(i)})^2 = \text{bias}^2 + \text{Variance} = (y_a - y_p)^2 + (\dots)$$

$\bar{g}(x)$: average over all lines



bias=0.21; var=1.69

bias=0.23; var=0.33

Polynomial Model

Want to fit a polynomial regression model

$$y = \theta_0 + \theta_1 x + \theta_2 x^2 + \cdots + \theta_d x^d + \epsilon$$

Let's rewrite it as:

$$y = \theta_0 + \theta_1 z_1 + \theta_2 z_2 + \cdots + \theta_d z_d + \epsilon = \mathbf{z}\boldsymbol{\theta}$$

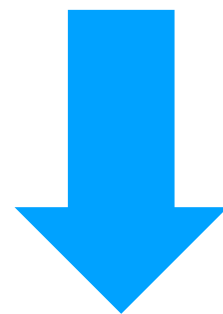
Regularizing is just constraining the weights (θ)

For example: let's do a **hard** constraining

$$y = \theta_0 + \theta_1 z_1 + \theta_2 z_2 + \cdots + \theta_d z_d$$

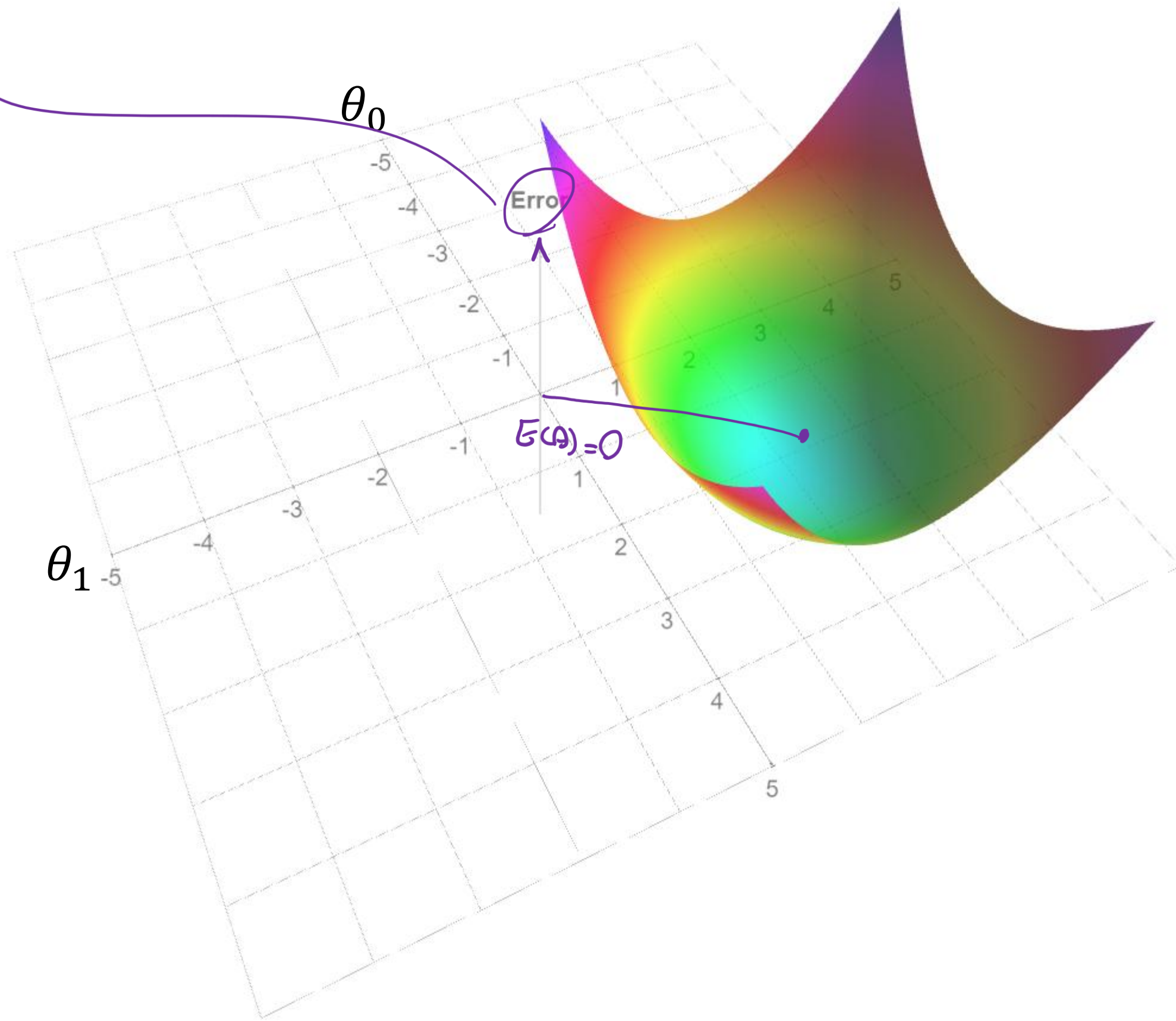
subject to

$$\theta_d = 0 \text{ for } d > 2$$

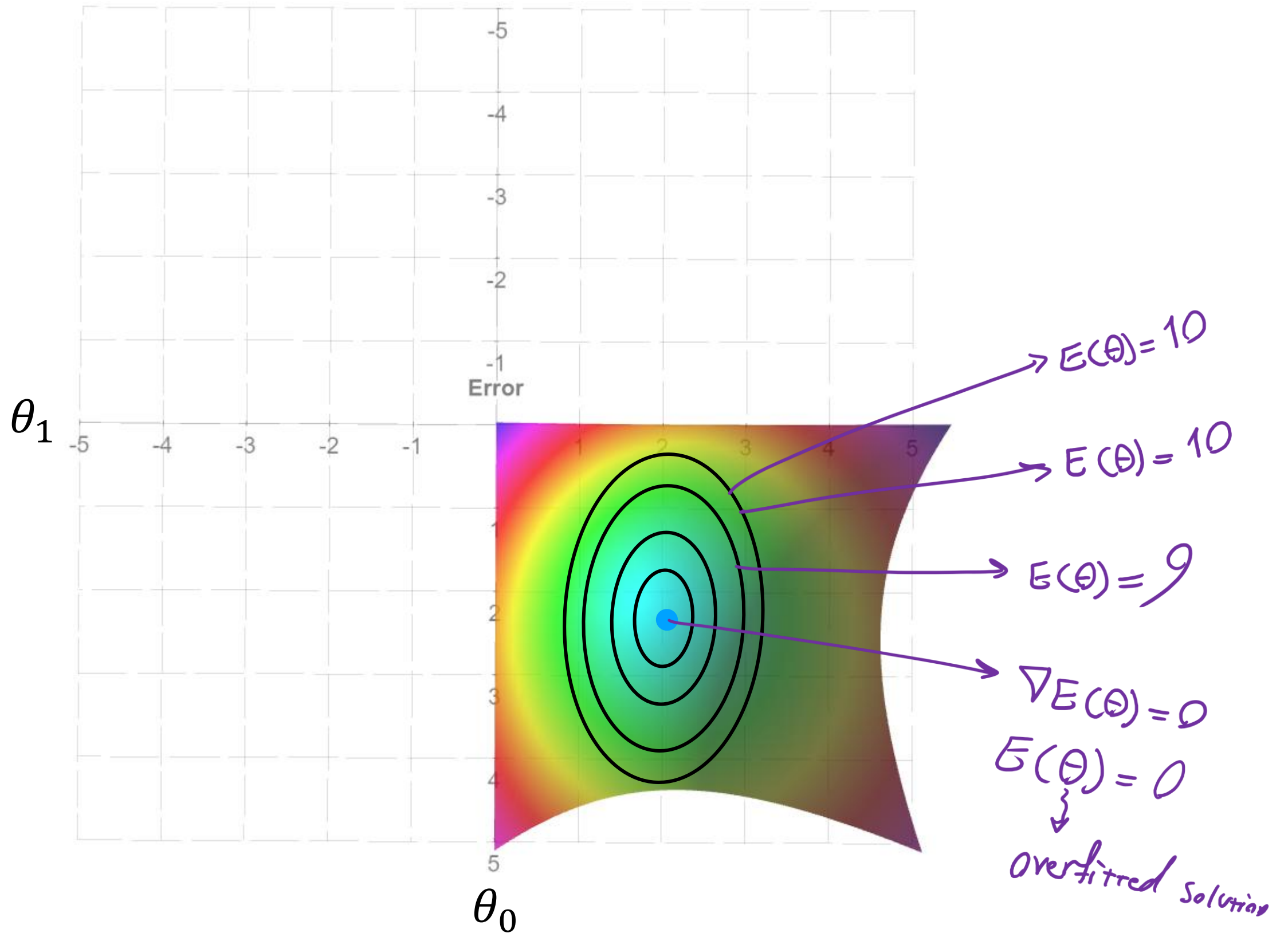


$$y = \theta_0 + \theta_1 z_1 + \theta_2 z_2 + 0 + \cdots + 0$$

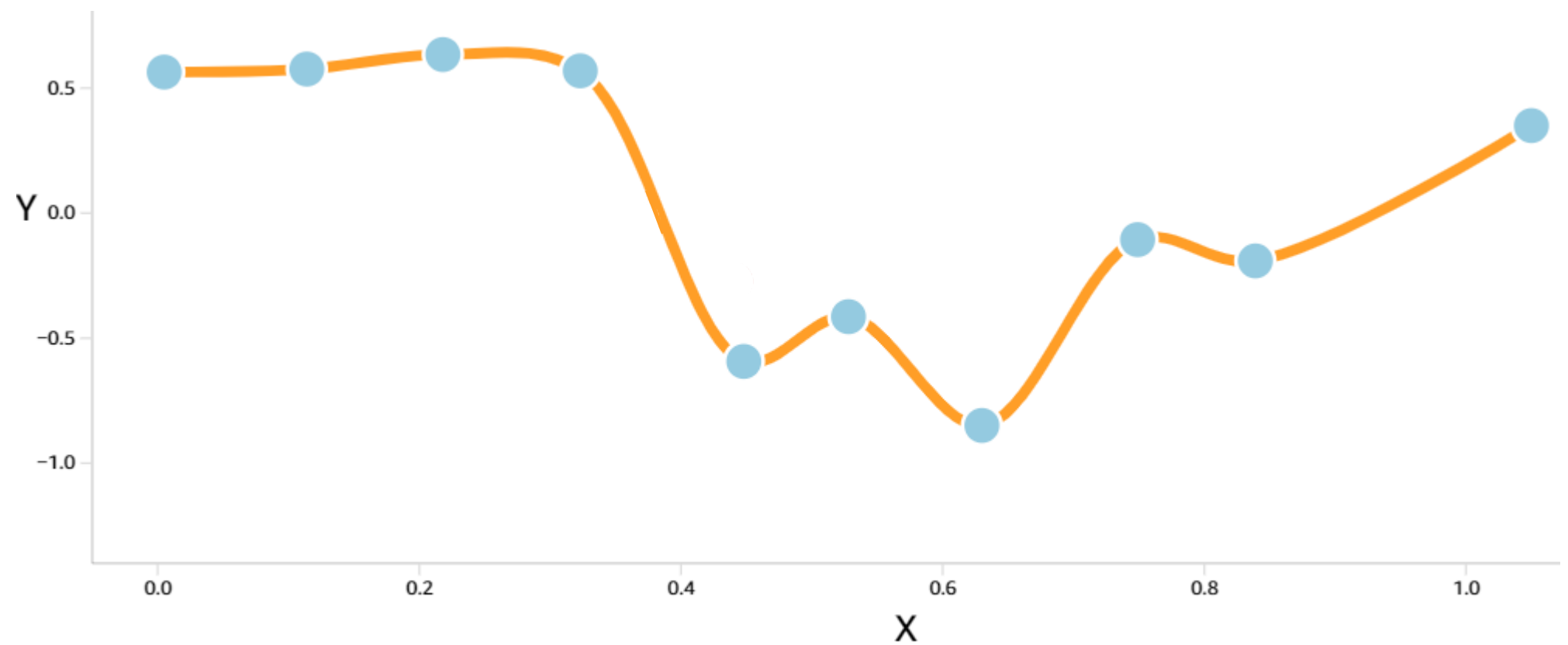
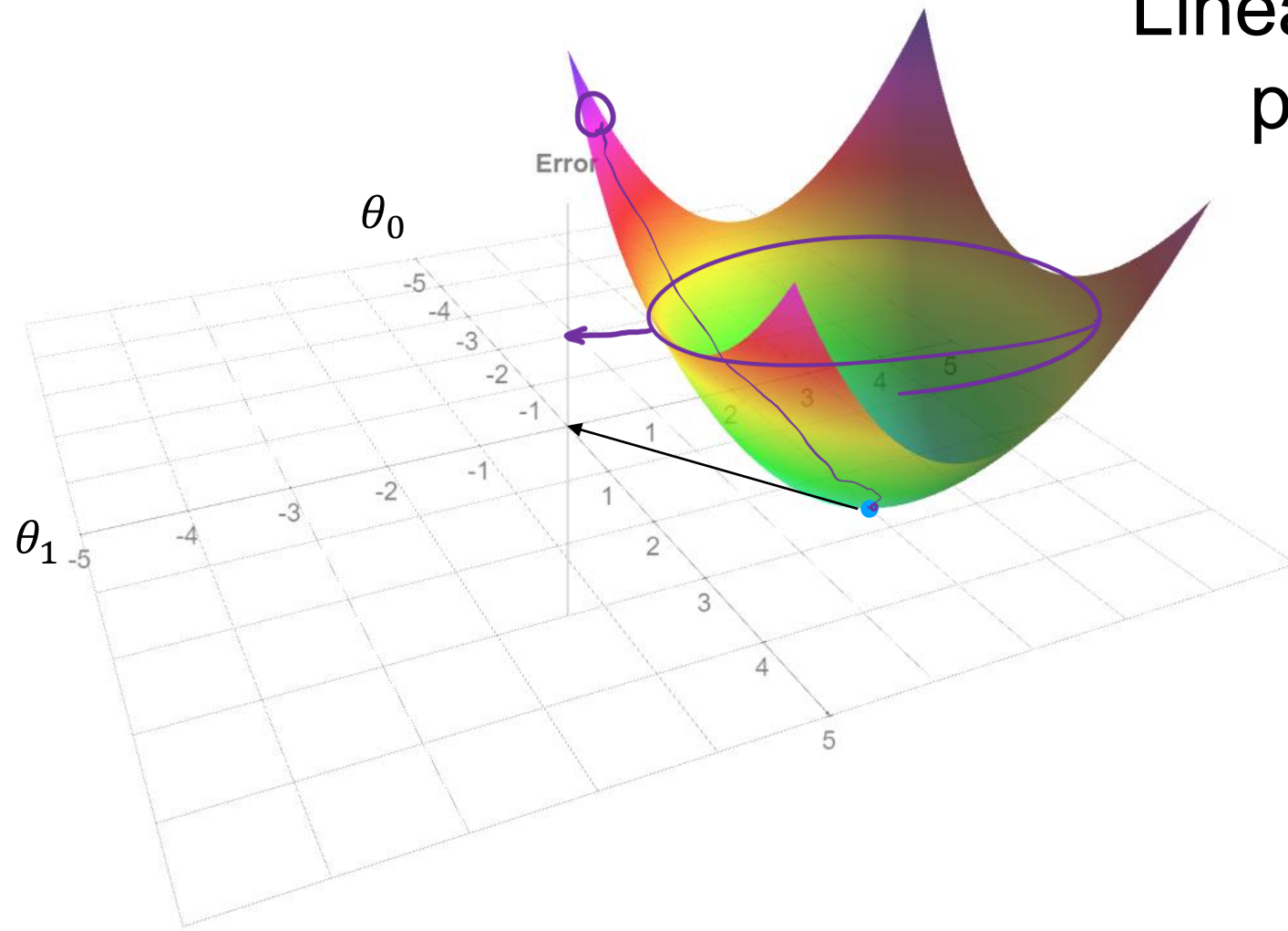
$$E(\theta) = \frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2 \rightsquigarrow \text{convex}$$

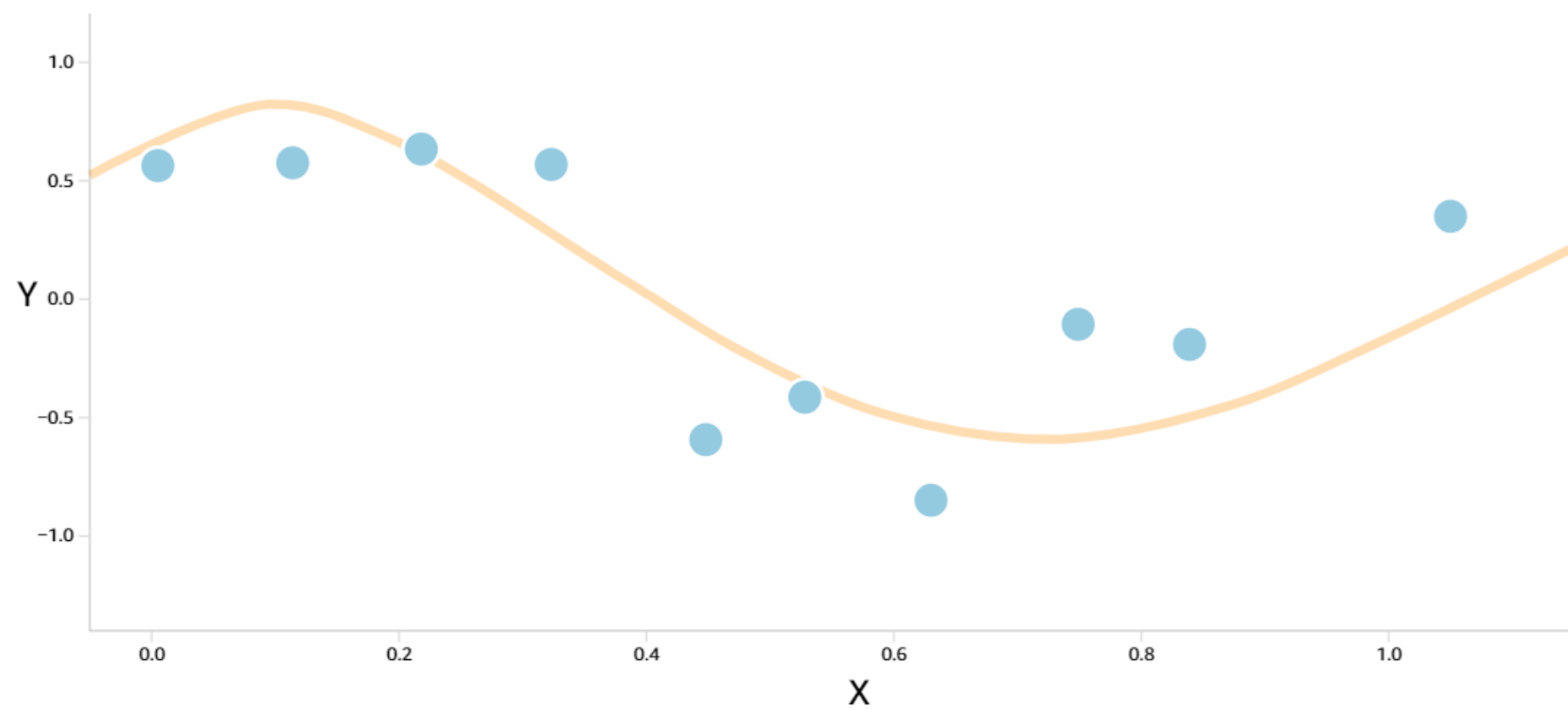
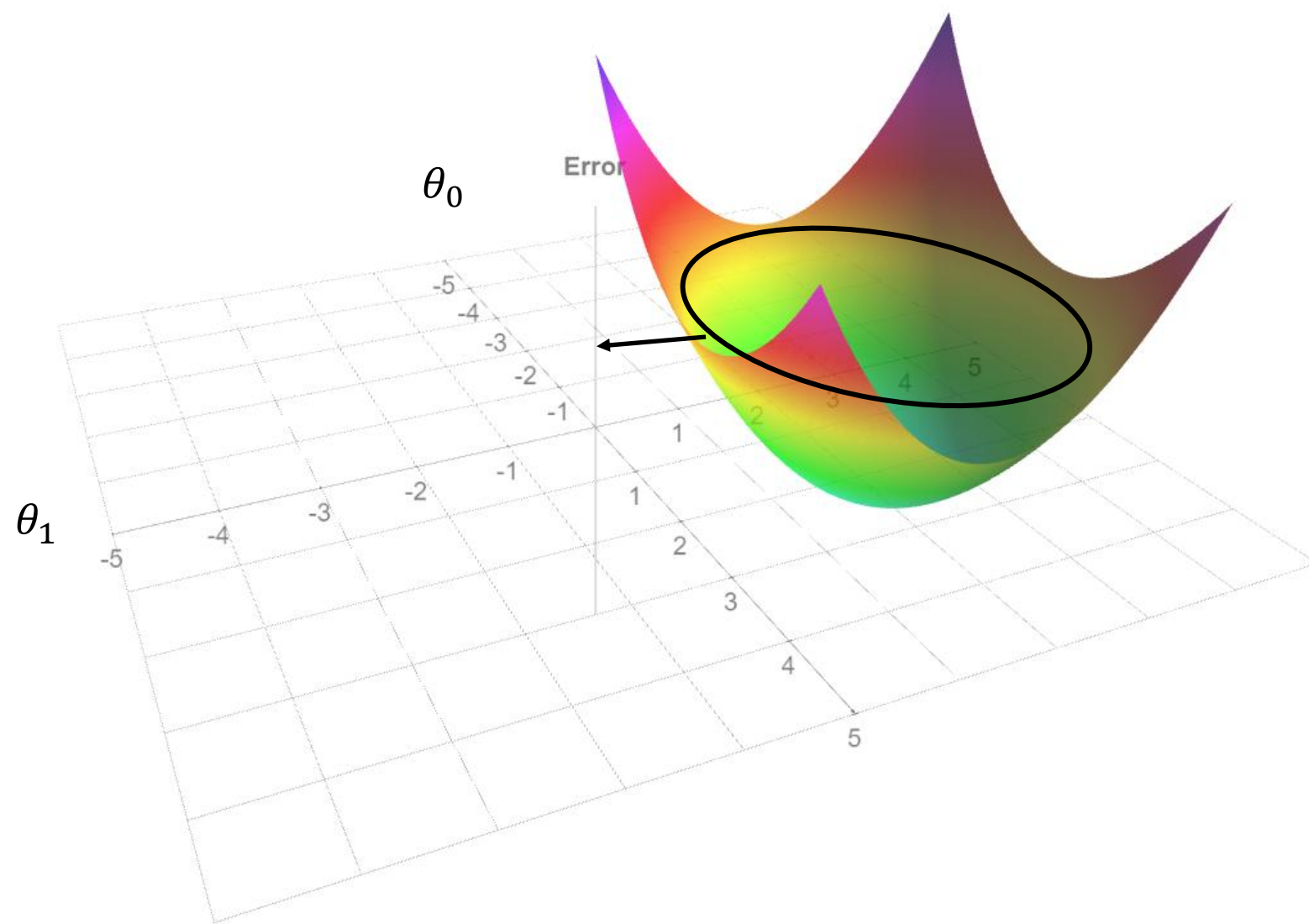


Project the same graph on x-y using contour plot



Linear regression with a very high polynomial degree solution

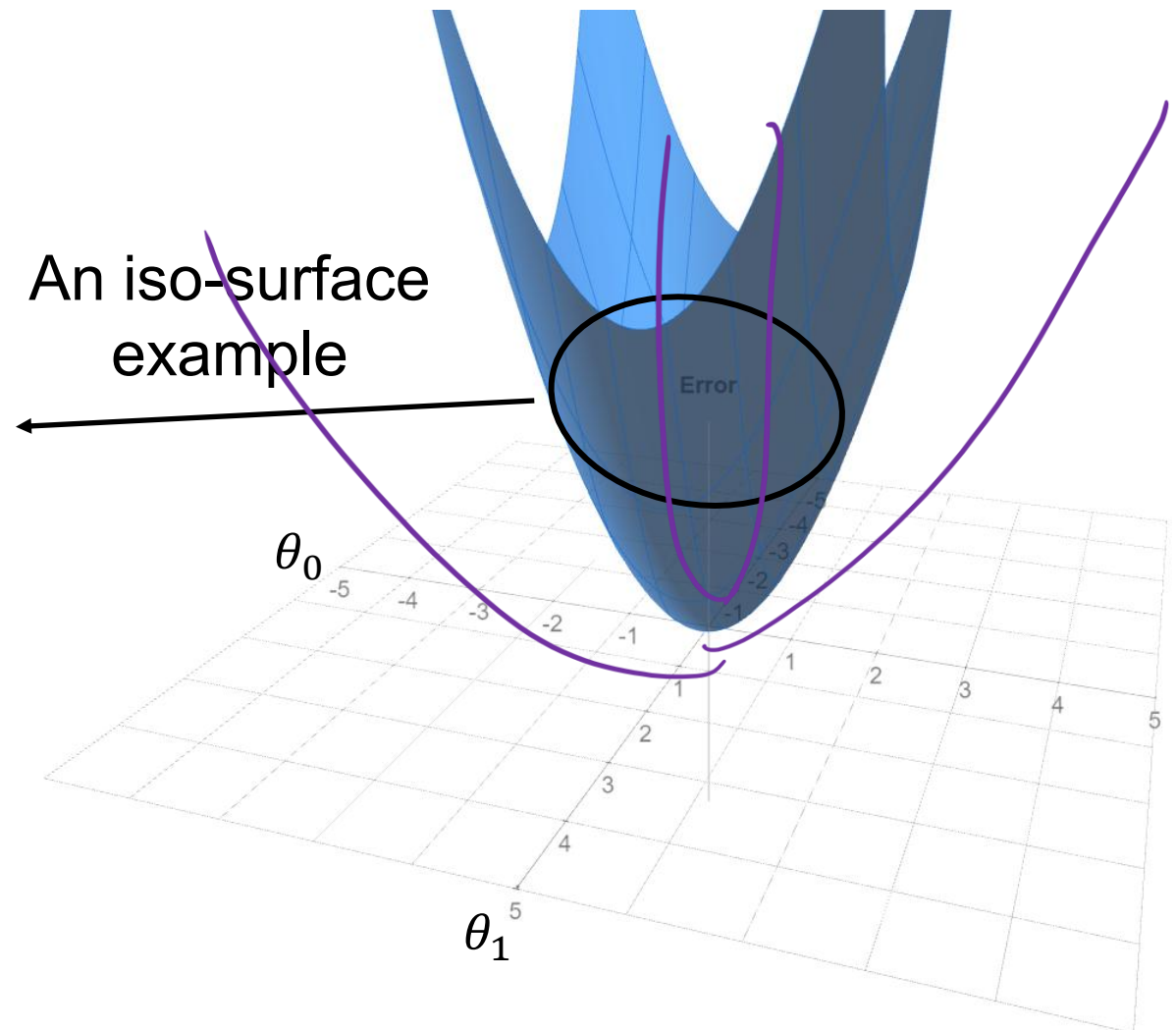




How can we get an optimal solution with a positive error for a model that overfits?

We need to introduce a constraint

$$\underline{g(\theta) = \theta_0^2 + \theta_1^2 = \theta^T \theta = \underline{C}}$$



Error function together with a
new introduced constraint

$$E(\theta)$$

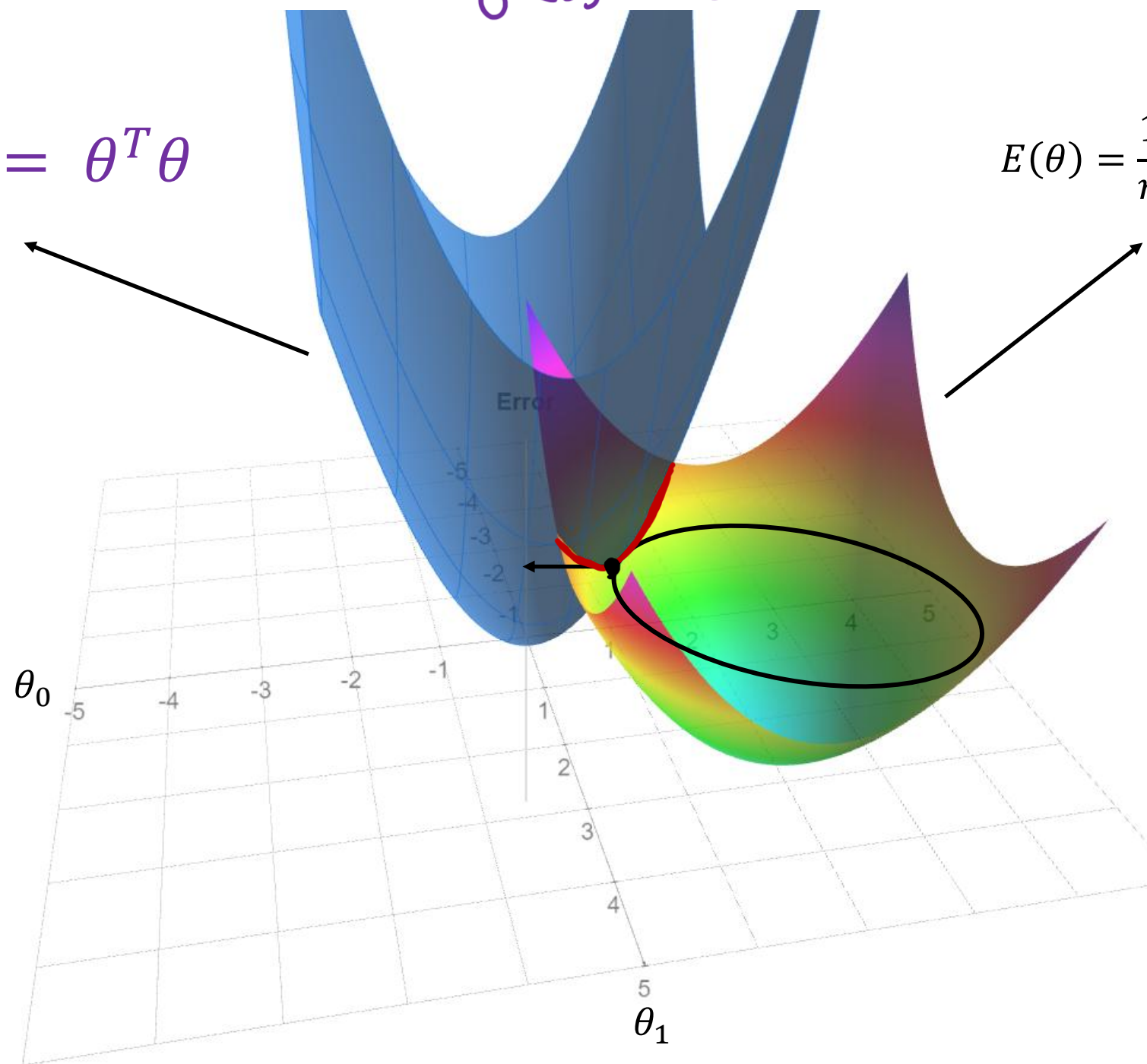
$$\text{s.t. } \theta^T \theta = c$$

$$g(\theta) = \theta^T \theta - c$$

$$\mathcal{L}(\theta, \lambda) = E(\theta) + \lambda(\theta^T \theta - c)$$

$$g(\theta) = \theta_0^2 + \theta_1^2 = \theta^T \theta$$

$$E(\theta) = \frac{1}{n} \sum_{i=1}^n (y^i - z_i \theta)^2$$



Let's define the Lagrange function

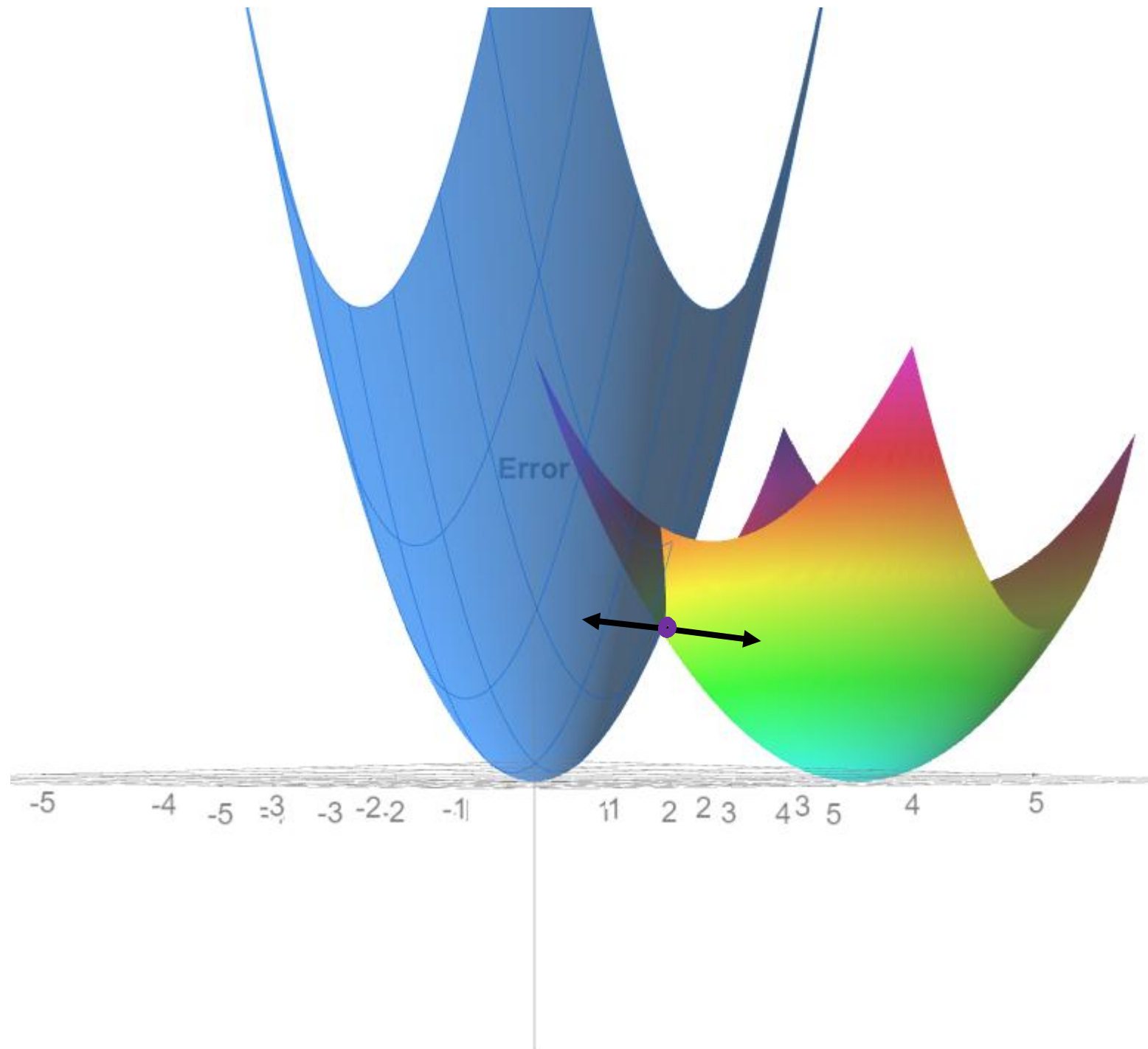
$$L(\theta, \lambda) = E(\theta) + \lambda g(\theta)$$

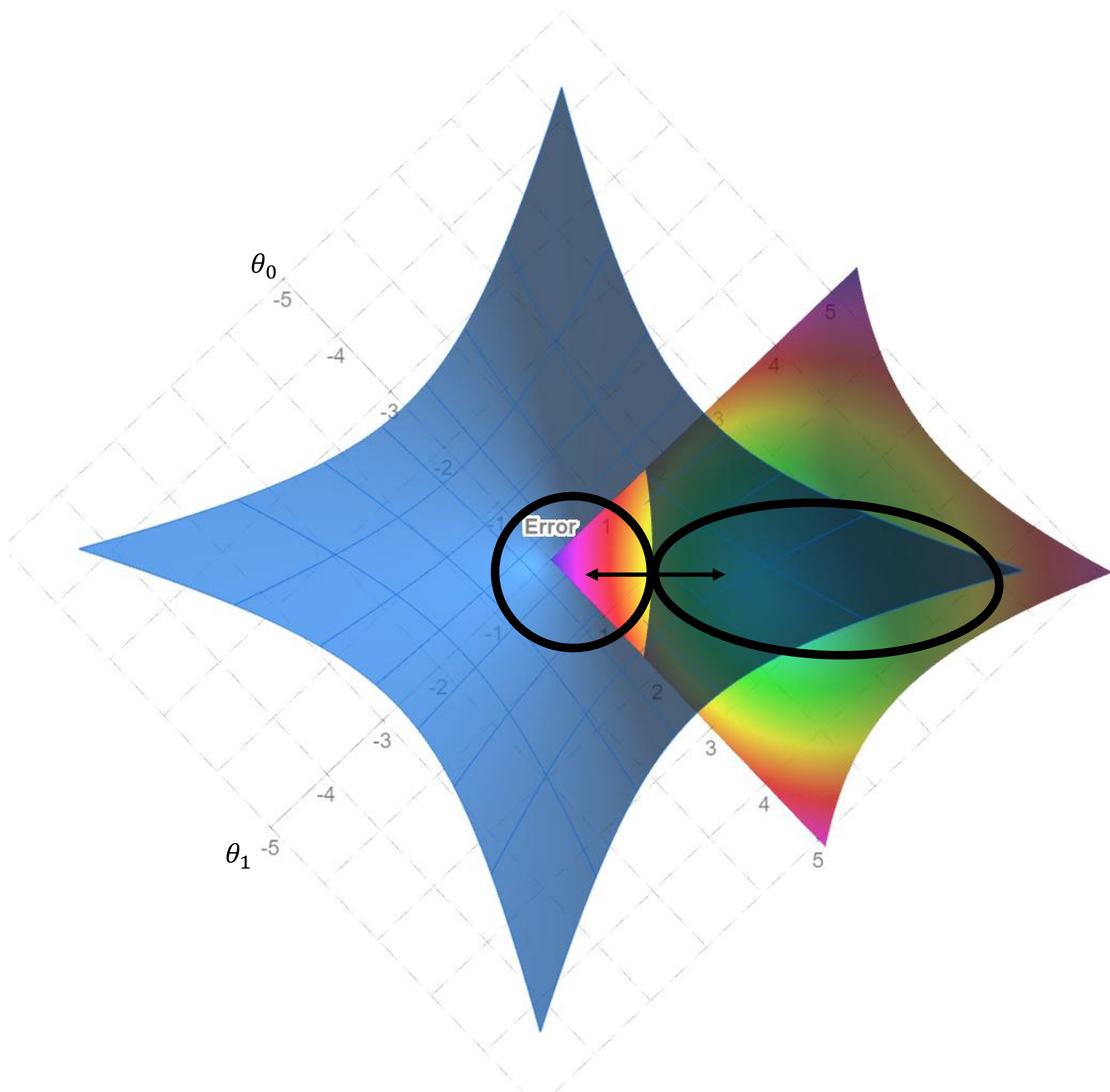
$$L(\theta, \lambda) = E(\theta) + \lambda(\theta^T \theta - c)$$

$$\nabla L(\theta, \lambda) = 0 \quad \nabla[E(\theta) + \lambda(\theta^T \theta - c)] = 0$$

$$\nabla[E(\theta)] + \lambda \nabla[\theta^T \theta] = 0$$

How to enforce the gradient of Lagrange function to be zero





$$E(\theta)$$

$$\text{s.t. } \theta^T \theta = C \Rightarrow g(\theta) = \theta^T \theta - C$$

w.r.t

$$\frac{\partial L}{\partial \theta} = 0 ; \quad \frac{\partial L}{\partial \lambda} = 0$$

$$L(\theta, \lambda) = E(\theta) + \lambda (\theta^T \theta - C)$$

What if λ was known to us?

$$L(\theta) = E(\theta) + \lambda \theta^T \theta - \lambda C$$

$\nearrow$ Constant
 $\rightarrow$ Constant

$$\underline{L(\theta)} = \underline{E(\theta)} + \left[\lambda \theta^T \theta \right] \rightarrow \text{Penalty term ; regularization term}$$

$\downarrow$
 $\theta^T \theta = C$

Let's calculate the gradients

Gradient of constraint $g(\theta)$

$$\nabla[\theta^T \theta] = 2\theta$$

$$\nabla[E(\theta)] + \lambda \nabla[\theta^T \theta] = 0$$

$$\nabla[E(\theta)] = -\lambda \nabla[\theta^T \theta]$$

$$\nabla E(\theta) = -2\lambda\theta$$

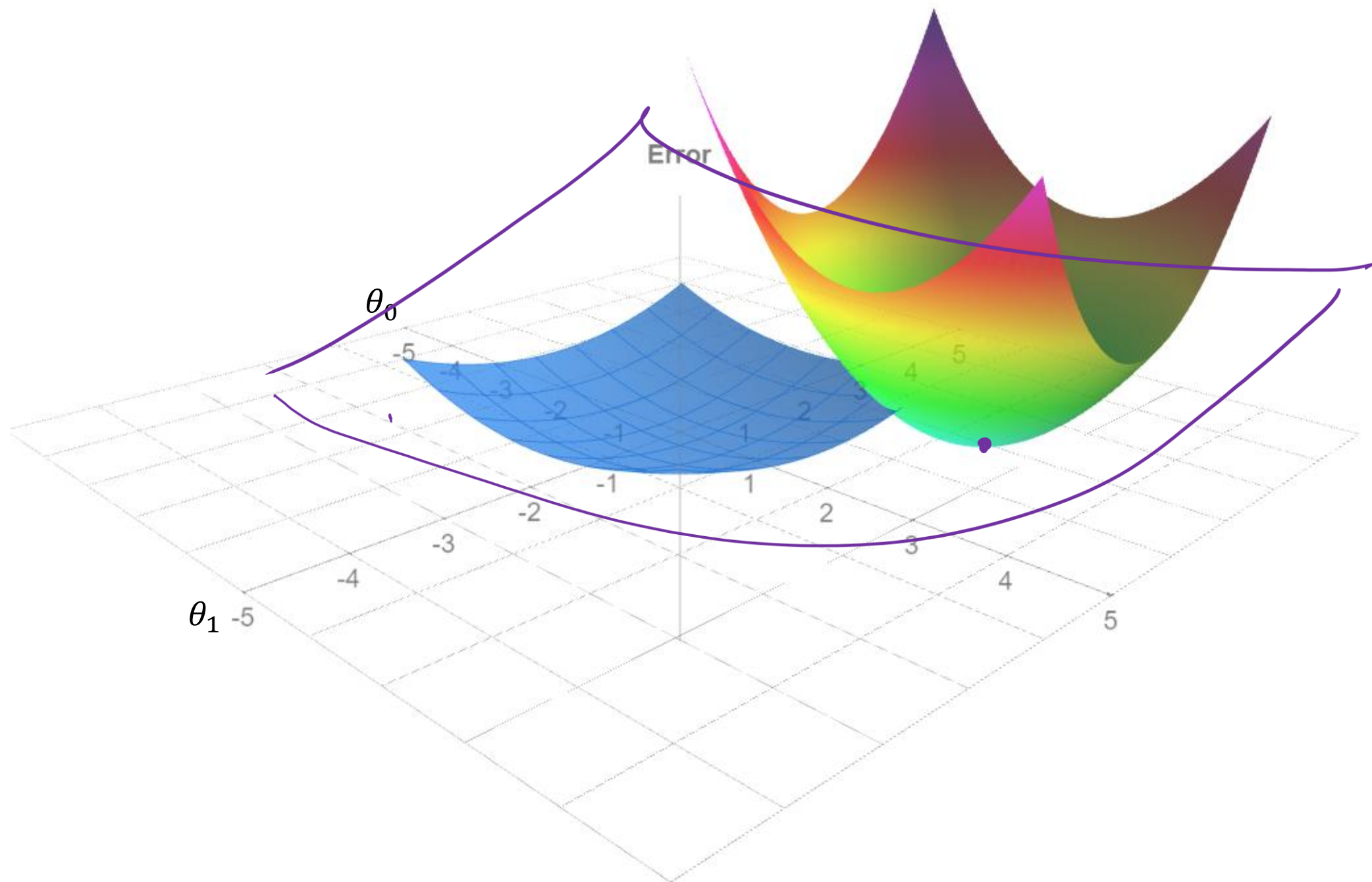
$$\nabla E(\theta) + 2\lambda\theta = 0$$

Let's do integration

$$E(\theta) + \lambda\theta^T \theta$$

The effect of low Lambda

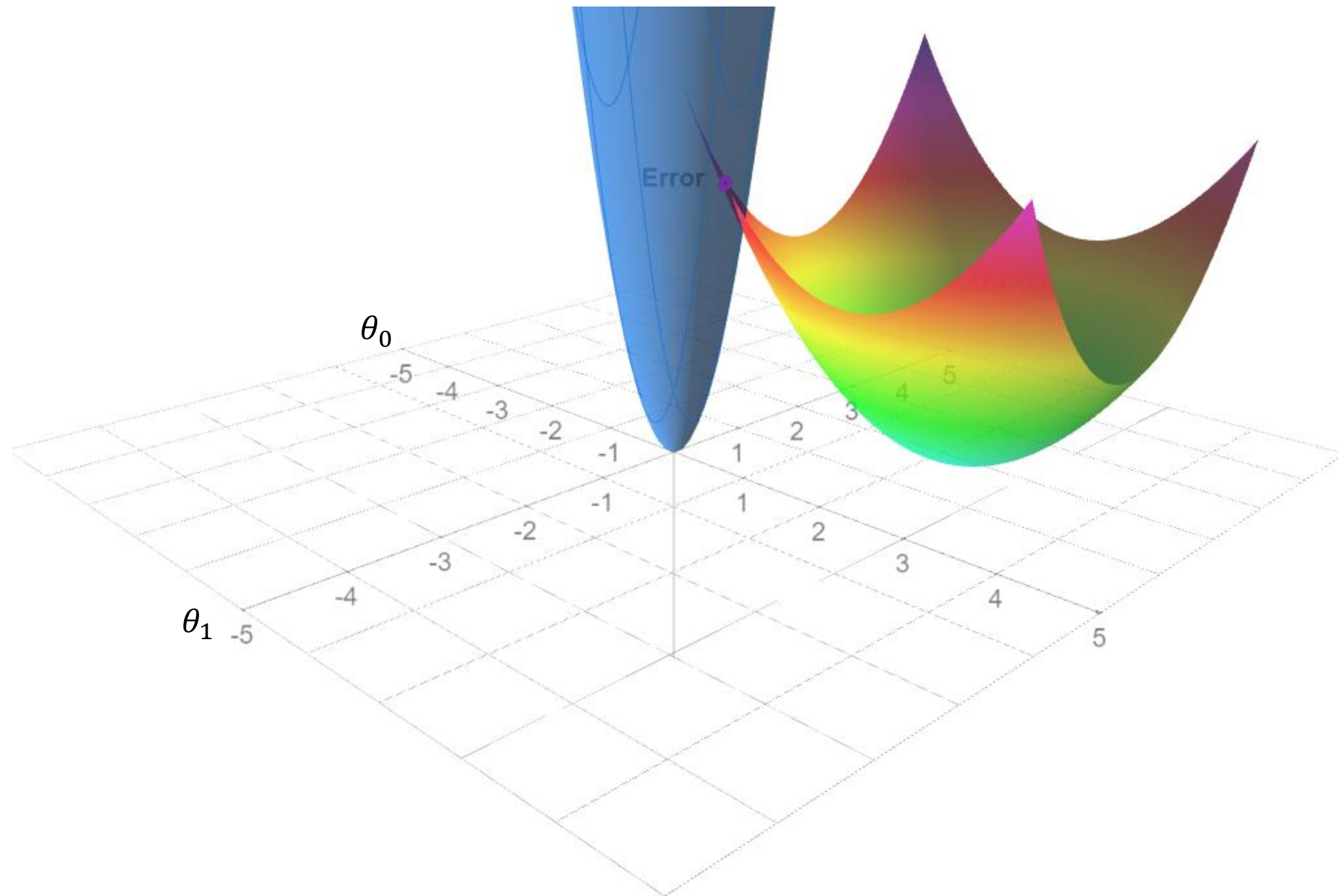
$$E(\theta) + \downarrow \frac{\lambda}{N} \theta^T \theta \quad \hookrightarrow \uparrow$$



The effect of high Lambda

$$E(\theta) + \frac{\lambda}{N} \theta^T \theta$$

Handwritten annotations: A purple arrow points up to λ , a purple arrow points down to $\theta^T \theta$, and a purple circle encircles $\theta^T \theta$.



Regularized Learning

Now we know Why this term
leads to the regularization of
parameters

Minimize $E(\theta) + \lambda \theta^T \theta$

Regularized Error

$$\tilde{E}(\theta) = \frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2 + \frac{\lambda}{2N} \|\theta\|_2^2$$

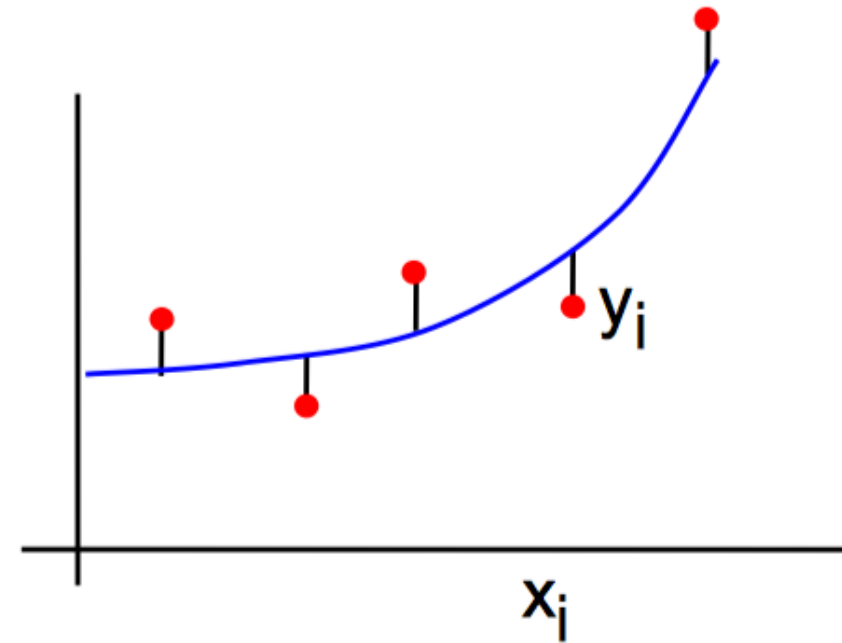
L2 Regularization term

Outline

- Overfitting and regularized learning
- Ridge regression 
- Lasso regression
- Determining regularization strength

Ridge Regression

$$\begin{aligned}\tilde{E}(\theta) \\ &= \frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2 + \frac{\lambda}{2N} \|\theta\|_2^2\end{aligned}$$



$$\theta_0 + \theta_1 z_1 + \theta_2 z_2 + \cdots + \theta_d z_d + \epsilon = \mathbf{z} \boldsymbol{\theta}$$

General form

$$\tilde{E}(\theta) = \frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2 + \lambda \|\theta\|_2^2$$

Matrix form

$$\tilde{E}(\theta) = \frac{1}{N} (y - z\theta)^T (y - z\theta) + \frac{\lambda}{2} \|\theta\|_2^2$$

$$\frac{\partial \tilde{E}(\theta)}{\partial \theta} = -z^T (y - z\theta) + \lambda \theta$$

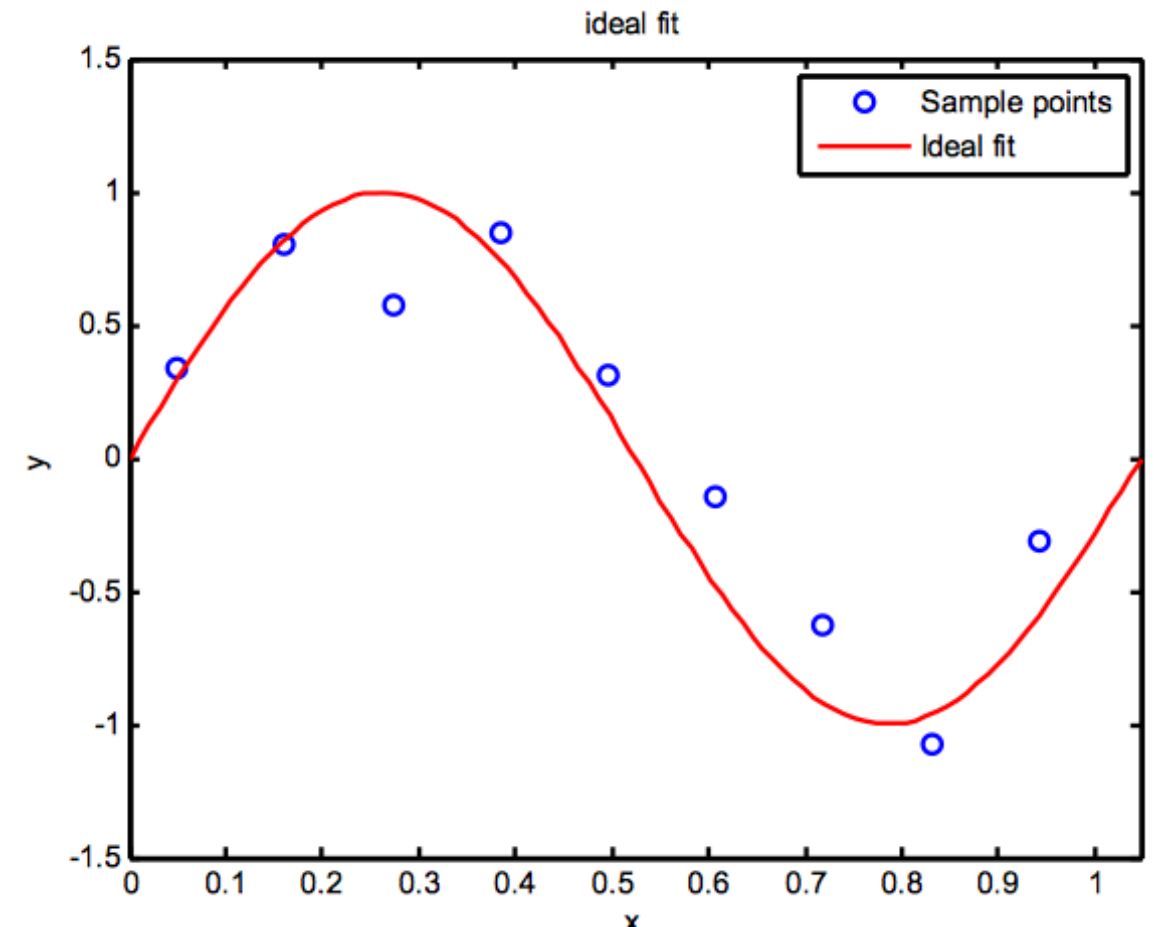
$$(z^T z + \lambda I) \theta = z^T y$$

$$\theta = (z^T z + \lambda I)^{-1} z^T y \approx \frac{1}{z^T z + \lambda} z^T y$$

$$\theta_{\text{overfitted}} = (z^T z)^{-1} z^T y$$

Ridge Regression Example

- The red curve is the true function (which is not a polynomial)
- The data points are samples from the curve with added noise in y.
- There is a choice in both the degree, D , of the basis functions used, and in the strength of the regularization

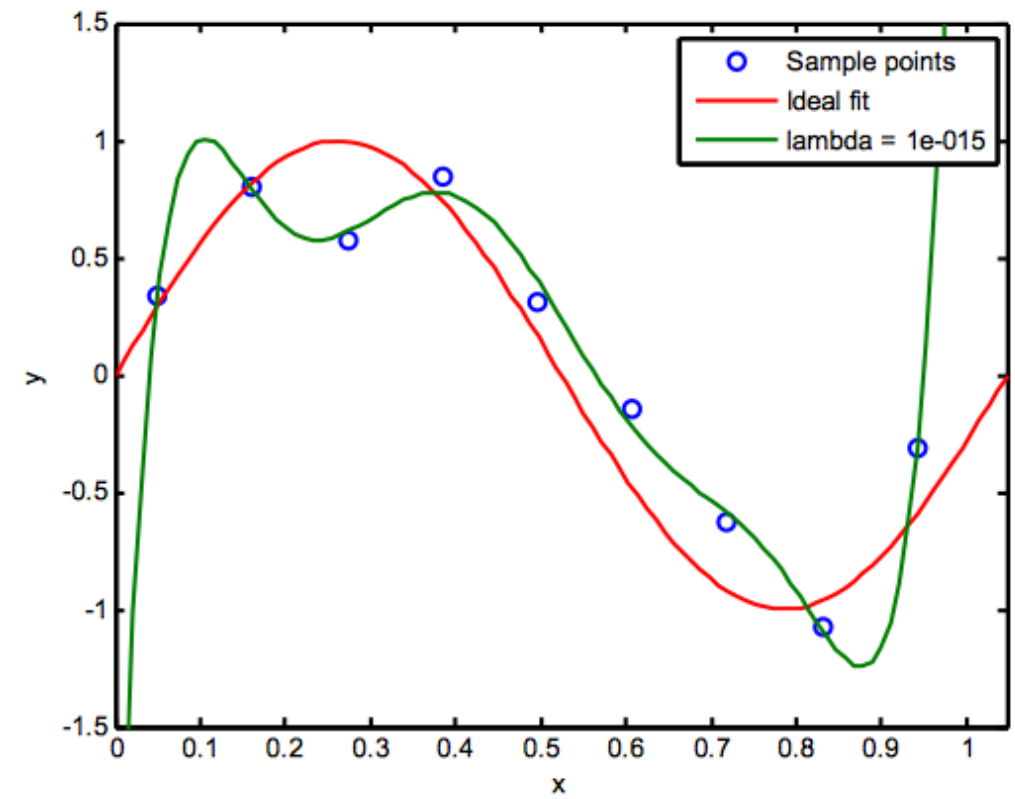
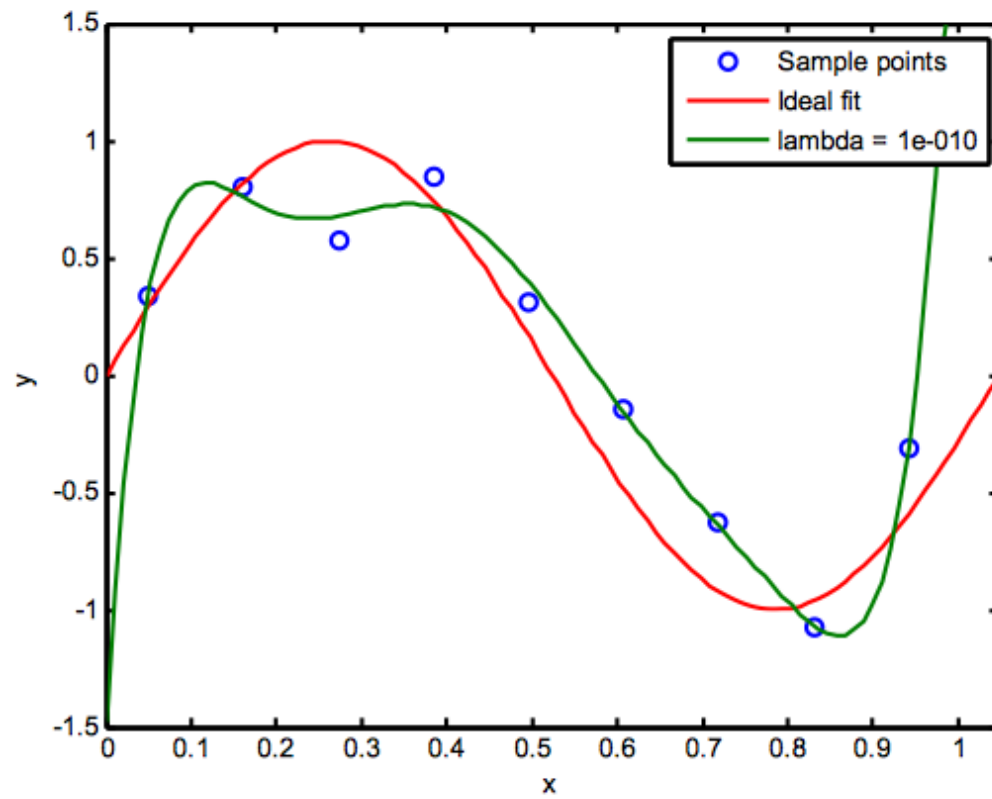
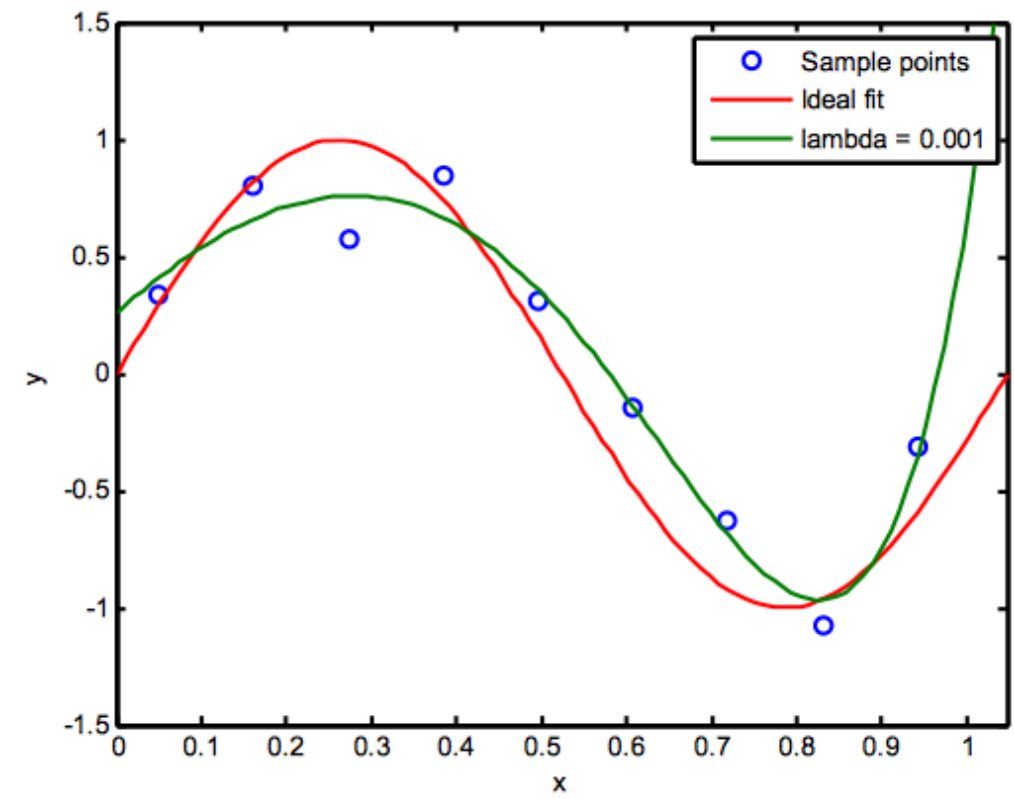
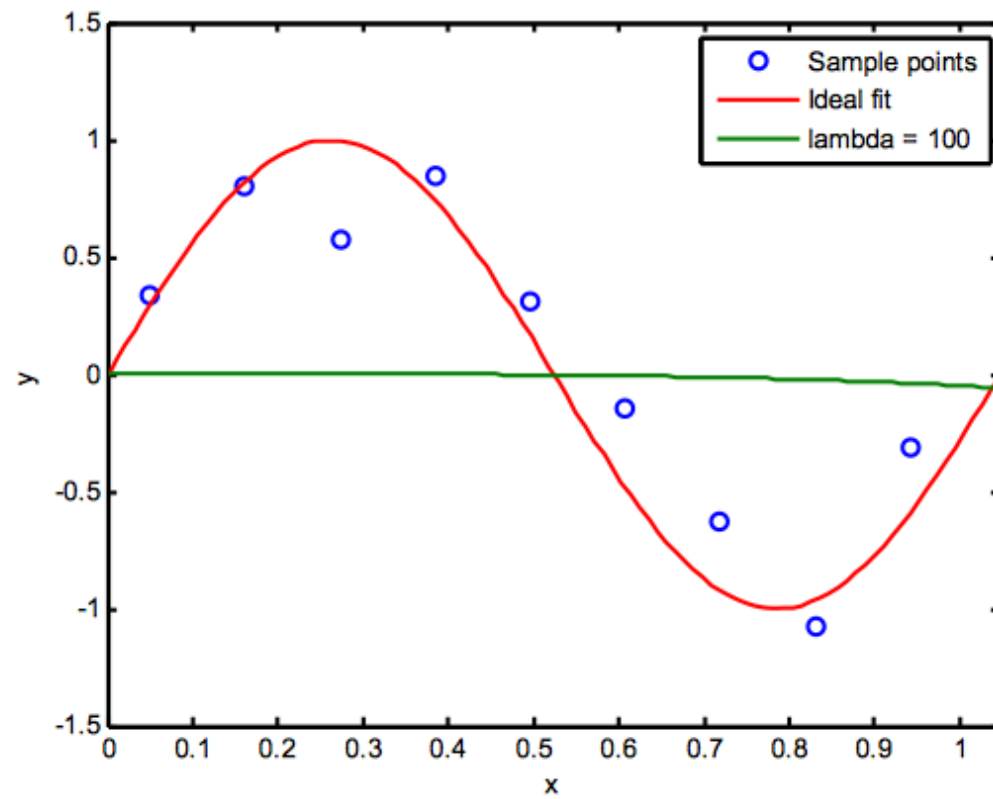


$$f(x, \theta) = z\theta$$

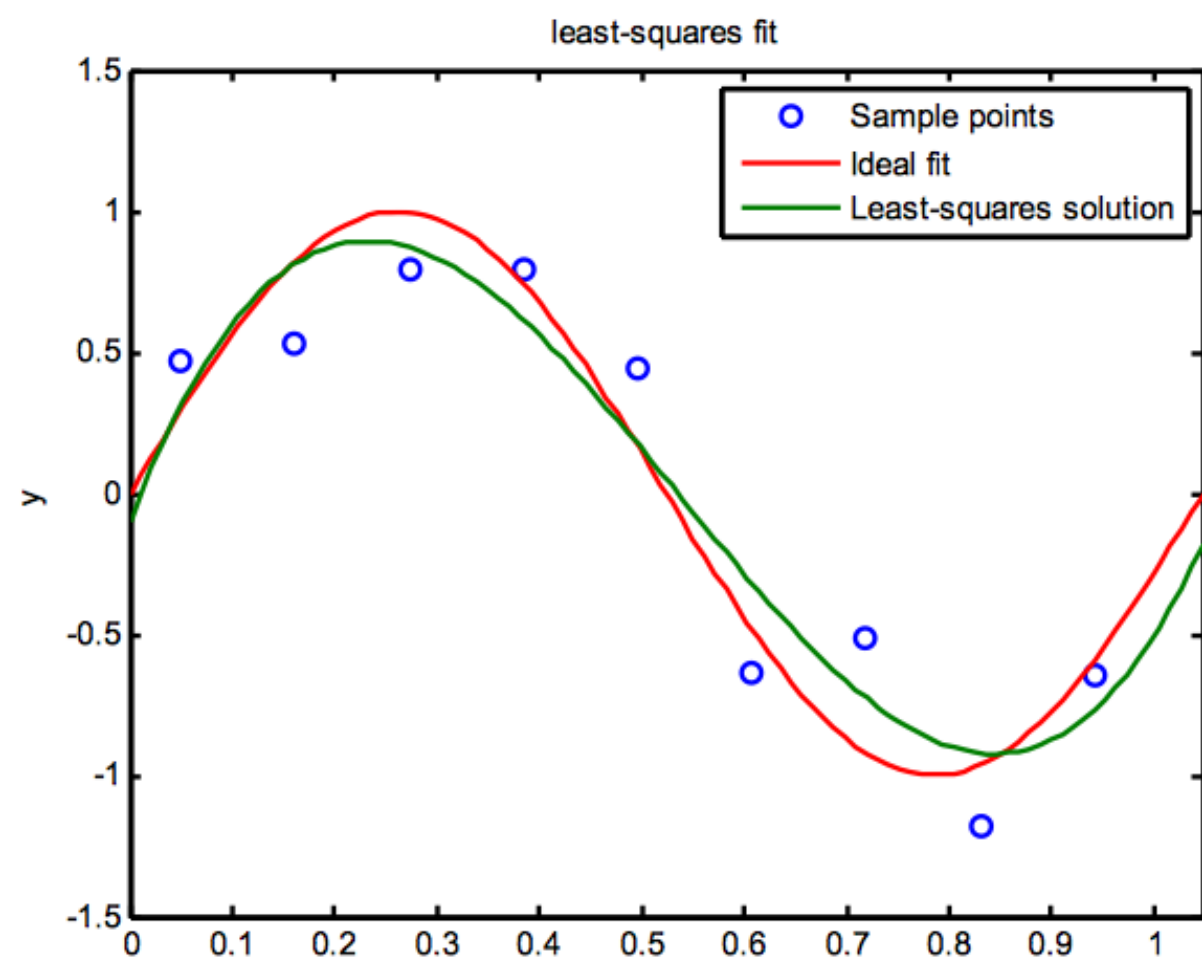
$$z: x \rightarrow z$$

$$\tilde{E}(\theta) = \frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}}\theta)^2 + \lambda \|\theta\|_2^2 \quad \theta \in \mathbb{R}^{D+1}$$

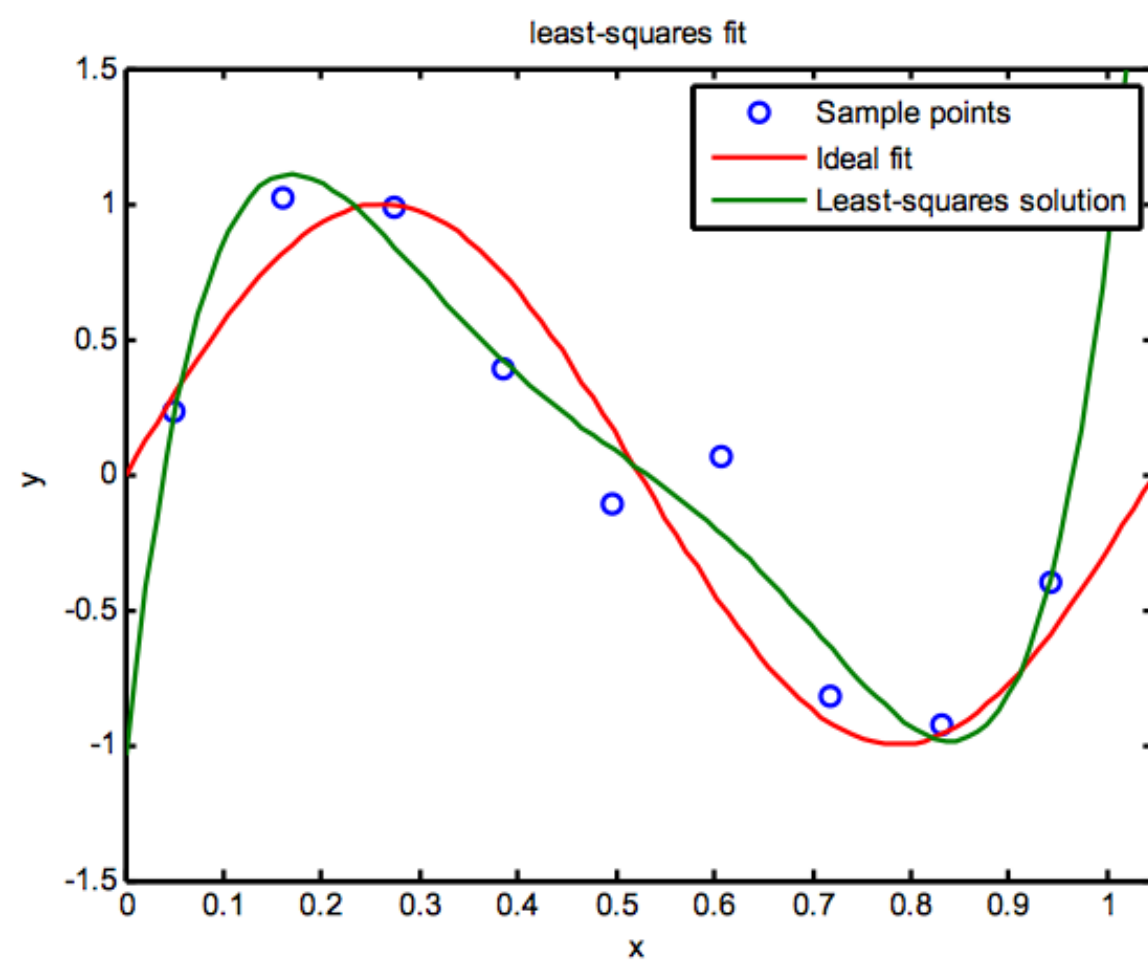
$N = 9$ samples, $D = 7$



$D = 3$



$D = 5$



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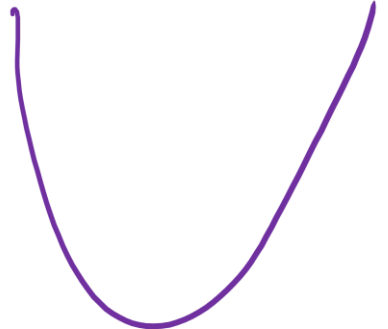
Dimensionality reduction → PCA $\leadsto X \rightarrow Z$ → LDA

Forward Feature selection

Backward " "

RF " "

Regularized Regression

$$\tilde{E}(\theta) = \frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2 + \lambda \|\theta\|_2^2$$


Squared loss/Error

$$\frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2$$

L2 Regularizer

$$\lambda \|\theta\|_2^2$$

Now let's look at another regularization choice.

The Lasso Regularization (L1 norm) and sparsity

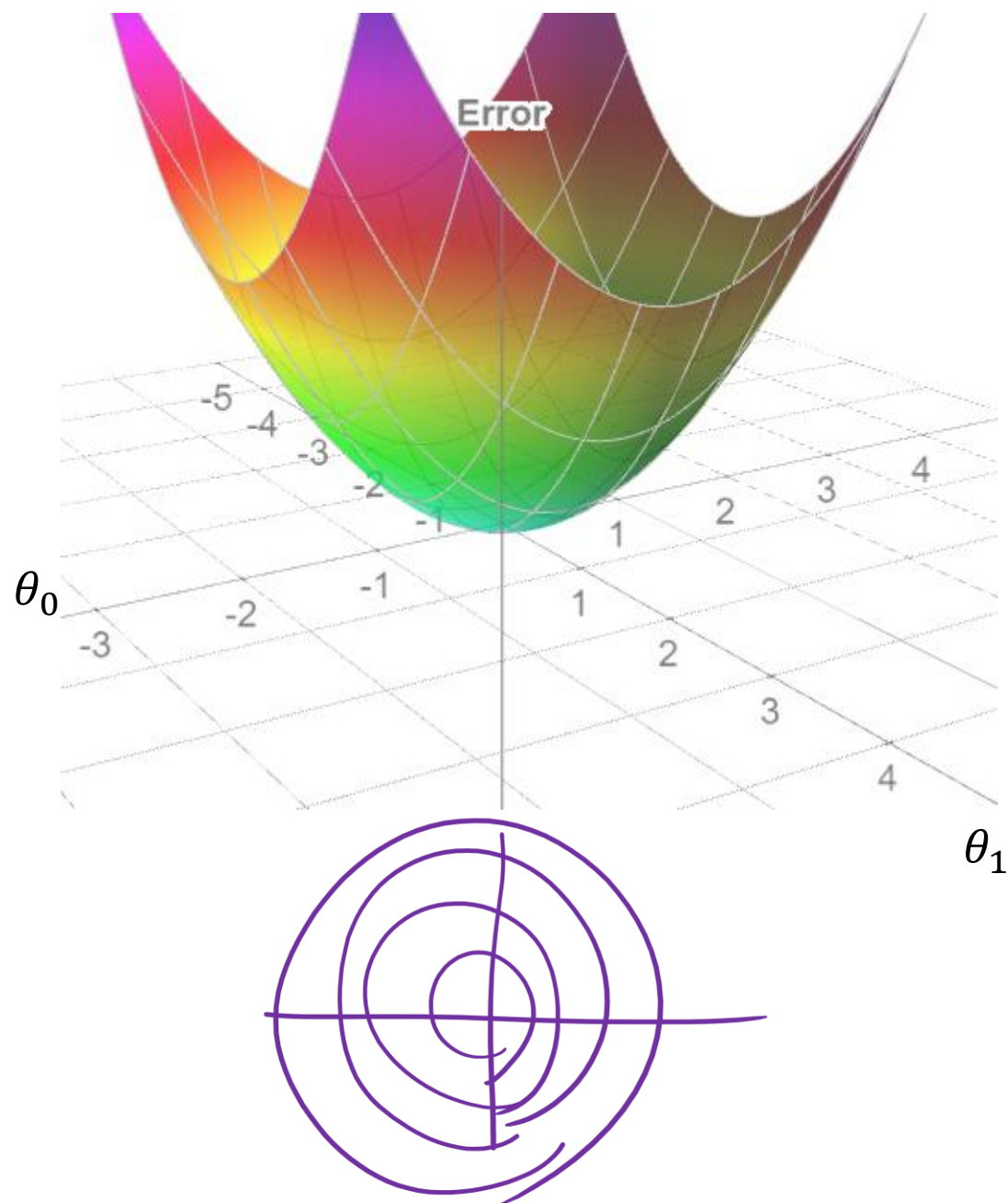
Lasso = **L**east Absolute **S**hrinkage and **S**election **O**perator

$$\tilde{E}(\theta) = \frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2 + \lambda \|\theta\|_1$$


L1 norm induces sparsity. This means that some of the weights become zero, and the feature contribution will be completely removed. L1 Regularizer could be used for feature selection

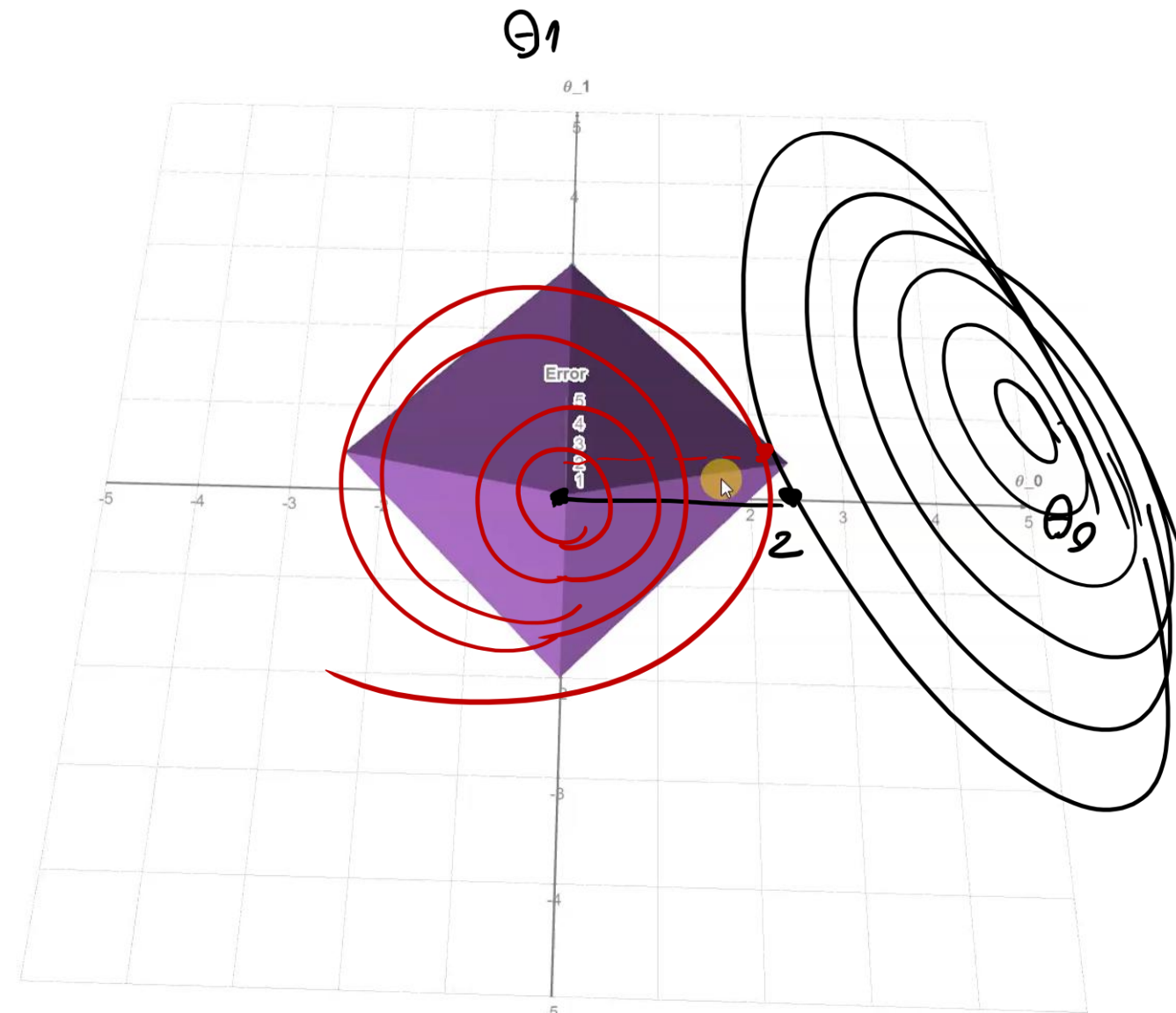
Ridge Regularizer

$$g(\theta) = \theta_0^2 + \theta_1^2 = \theta^T \theta$$



Lasso Regularizer

$$g(\theta) = \theta_0 + \theta_1 = \theta$$



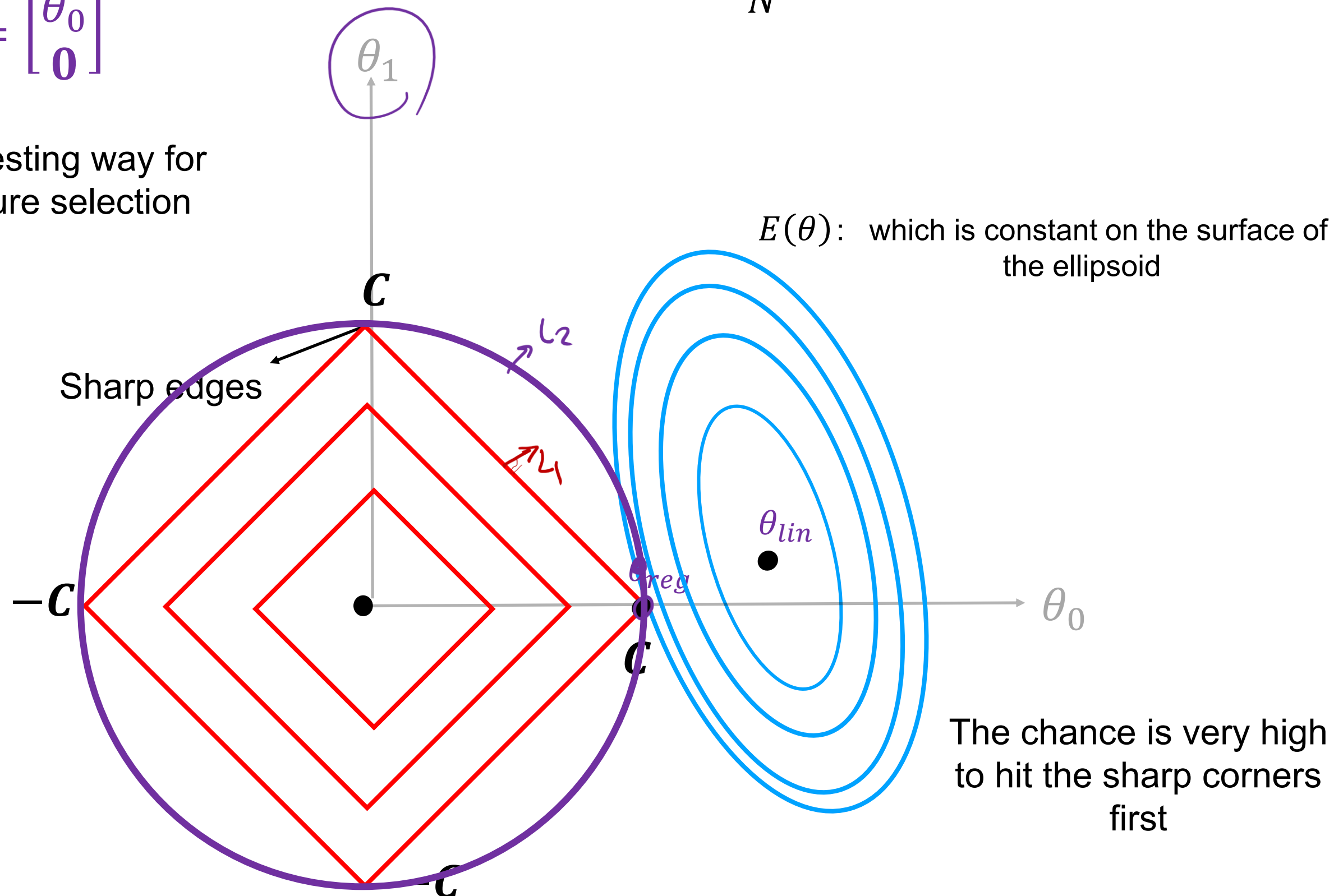
[Animation](#)

Let's say we have two parameters (θ_0 and θ_1)

$$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix}$$

$$\text{Min } E(\theta) = \frac{1}{N} (z\theta - y)^T (z\theta - y) + \lambda \|\theta\|_1$$

Interesting way for
feature selection



Ridge versus Lasso

Ridge

$$\tilde{E}(\theta) = \frac{1}{N} (y - z\theta)^T (y - z\theta) + \lambda \|\theta\|_2^2$$

It is a convex model

Both mean squared error and L2 regularizer are differentiable.

We can get a closed form solution

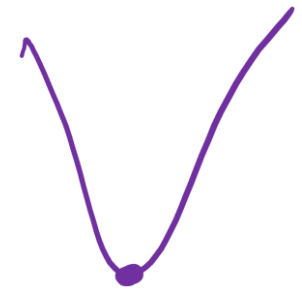
Lasso

$$\tilde{E}(\theta) = \frac{1}{N} (y - z\theta)^T (y - z\theta) + \lambda \|\theta\|_1$$

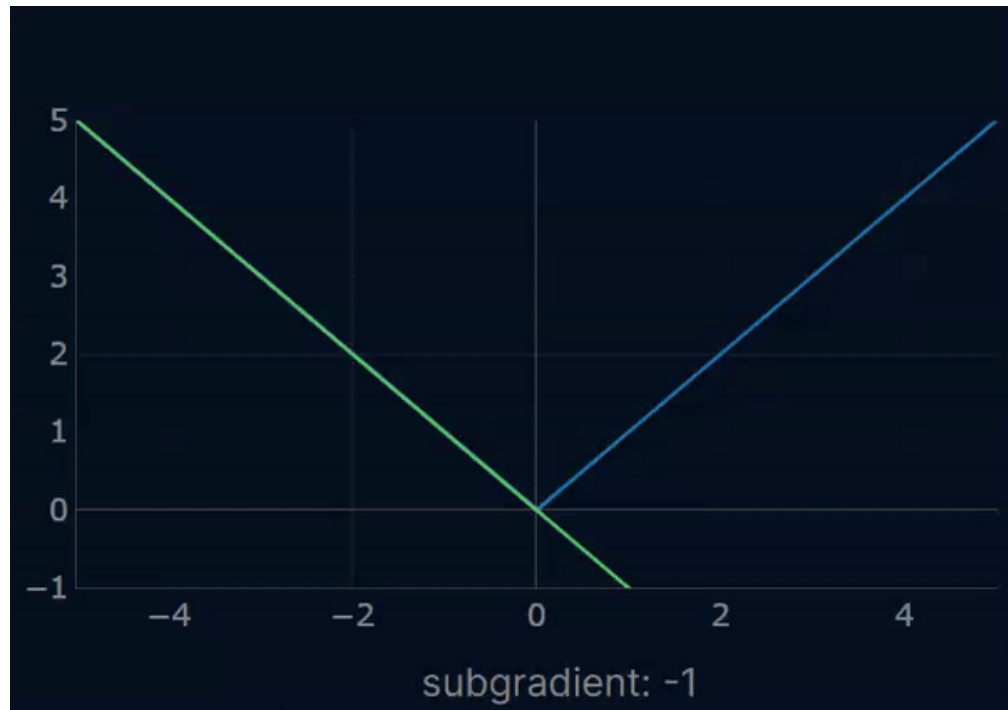
It is a convex model

L1 regularizer is NOT differentiable.

We can **NOT** get a closed form solution



Sub-gradient Descend in Lasso



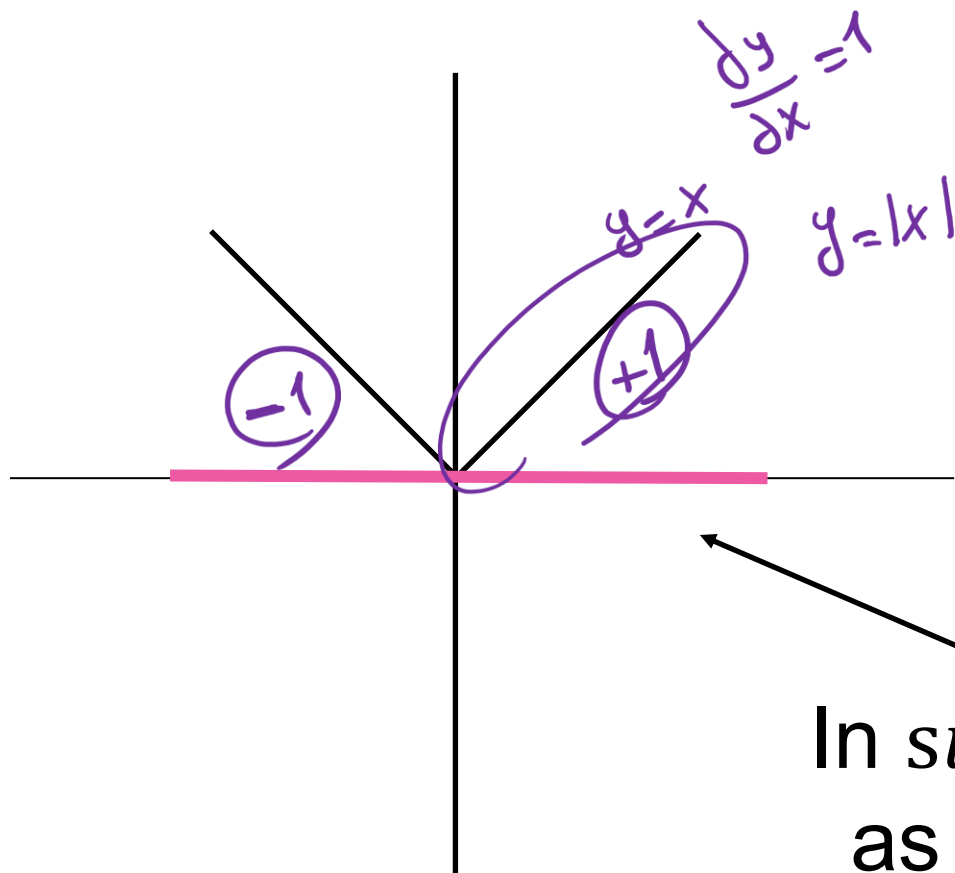
$$\tilde{E}(\theta) = \frac{1}{N} (y - z\theta)^T (y - z\theta) + \lambda \|\theta\|_1$$

$$\frac{\partial \tilde{E}(\theta)}{\partial \theta} = -z^T (y - z\theta) + \frac{\partial (\lambda \|\theta\|_1)}{\partial \theta}$$

$$\theta \leftarrow \theta - \alpha \frac{\partial \tilde{E}(\theta)}{\partial \theta}$$

Using Sub-gradient

$$\frac{\partial \tilde{E}(\theta)}{\partial \theta} = -z^T (y - z\theta) + \lambda \text{sign}(\theta)$$



In *sign* function, we use this sub-gradient line as our under-estimator (below our function)

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$$\theta = (Z^T Z + \lambda I)^{-1} Z^T y$$

$d+1 \times 1$

$$\theta = (Z^T Z)^{-1} Z^T y$$

$$\Theta = (Z^T Z + \delta I)^{-1} Z^T y$$

Leave-One-Out Cross Validation

$$\delta = 0.01$$

$$\delta = 0.02$$

⋮

$$\delta = 0.1$$

For every $i = 1, \dots, n$:

$$E(\Theta) = L(\Theta) = \frac{1}{N} \sum_{i=1}^N (y_a^{(i)} - z^{(i)} \Theta)^2$$

- ▶ train the model on every point except i ,
- ▶ compute the test error on the held out point.

Average the test errors.

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i^{(-i)})^2$$

$$\frac{1}{N} \sum_{i=1}^N E_i(\Theta)$$



Training



$$E_1(\Theta) = (y_a^{(1)} - z^{(1)} \Theta)^2$$

$$E_2(\Theta)$$

$$E_n(\Theta)$$

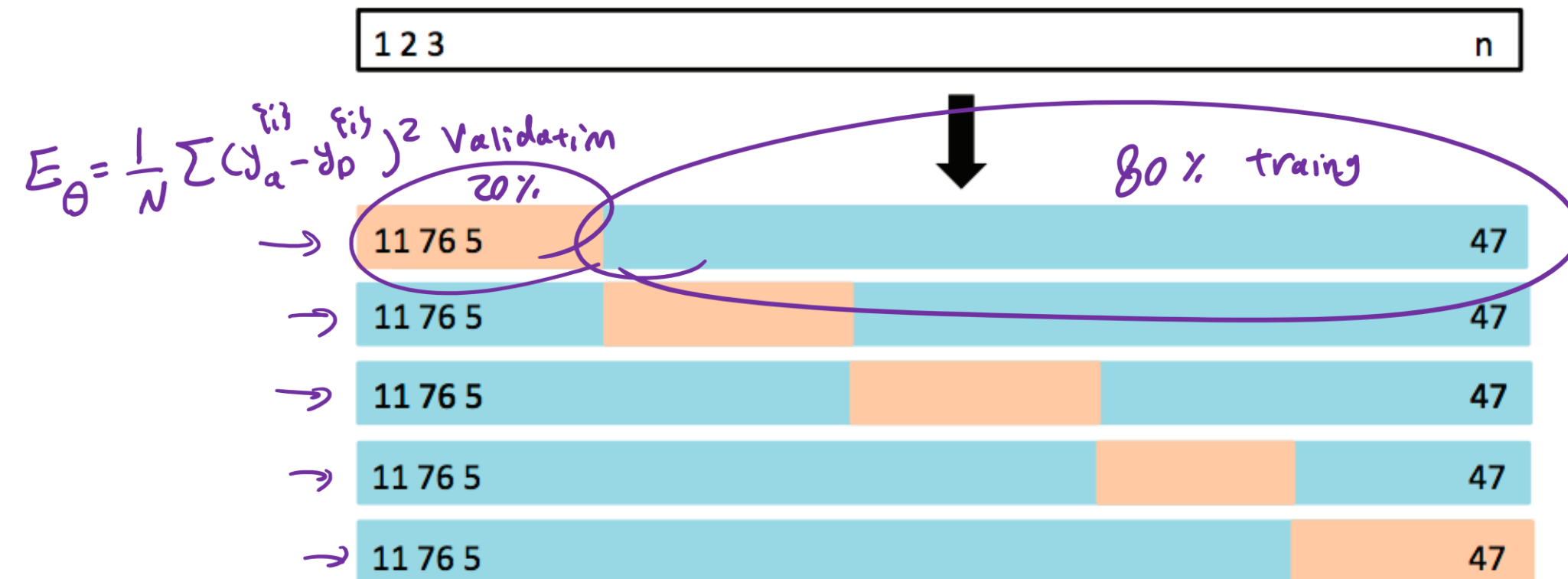
K-Fold Cross Validation

Split the data into k subsets or *folds*.

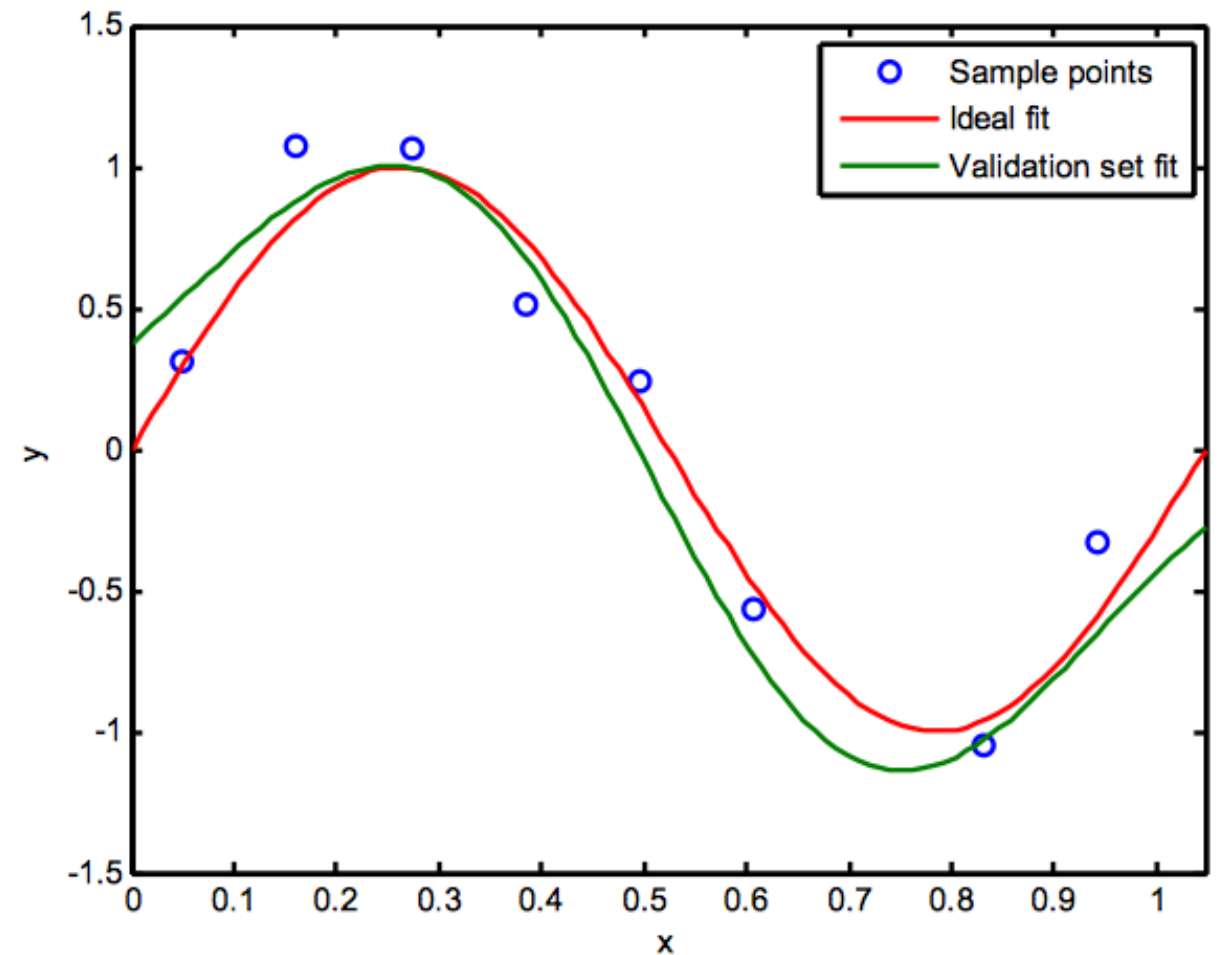
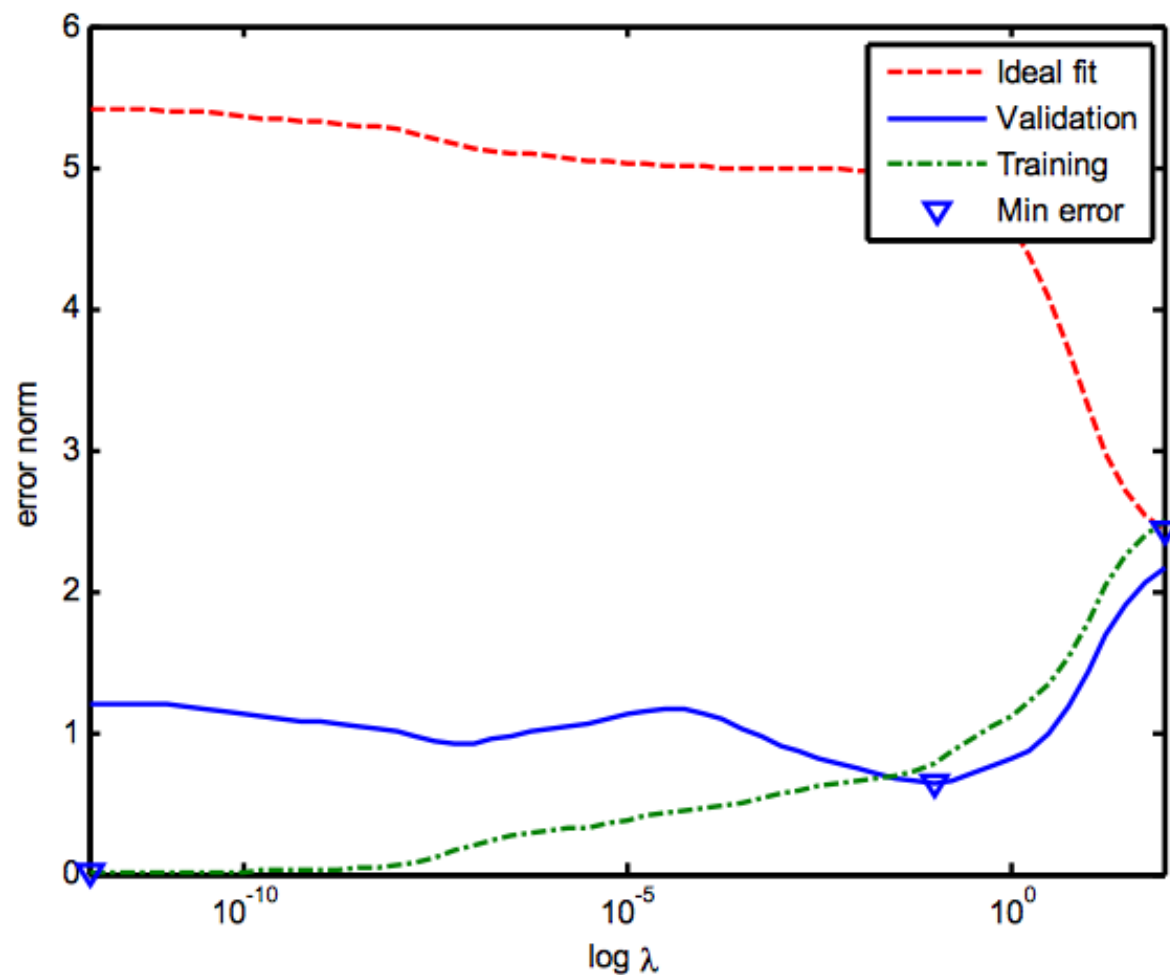
For every $i = 1, \dots, k$:

- ▶ train the model on every fold except the i th fold,
- ▶ compute the test error on the i th fold.

Average the test errors.



Choosing λ Using Validation Dataset



Pick up the lambda with the lowest
mean value of rmse calculated by
Cross Validation approach

Take-Home Messages

- What is overfitting
- What is regularization
- How does Ridge regression work
- Sparsity properties of Lasso regression
- How to choose the regularization coefficient λ